

REVIEW

Open Access



Ensuring academic integrity through automated online exam proctoring a decade long systematic review

Manit Malhotra^{1*} and Indu Chhabra¹

*Correspondence:

Manit Malhotra
manitmalhotra@rediffmail.com
¹Department of Computer Science
and Applications, Panjab University,
Chandigarh, India

Abstract

Academic cheating in online examinations jeopardizes academic integrity and necessitates prompt intervention to uphold the credibility of digital assessments. Traditional proctoring methods are increasingly inadequate in detecting students' cheating behaviors, especially in the growing landscape of digital education. This study highlights the pressing need for advanced automated solutions to ensure fairness in online examinations, with Artificial Intelligence Proctoring Systems (AIPS) offering a transformative approach. The research focuses on the application of Machine Learning (ML) and Deep Learning (DL) techniques in developing reliable, Automated Proctoring Systems (APS). This research synthesizes 80 peer-reviewed articles published from 2014–2024 in accordance with predetermined inclusion/exclusion criteria to assess the performance, reliability, and limitations of AI-based proctoring systems (including the hardware). In contrast to earlier reviews, this article provides a more thorough synthesis across multiple dimensions. The findings reveal that advanced ML and DL techniques, such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), better detect cheating by analyzing visual cues, including eye movements, head posture, and facial expressions, as compared to traditional techniques. Key evaluation metrics such as precision, recall, F1-score, specificity, and sensitivity are highlighted where available. The findings advocate for the integration of Internet of Things (IoT) and biometric technologies to augment the security, scalability, and operational efficiency of online proctoring systems. This comparative synthesis informs the development of a unified, robust monitoring framework capable of redefining academic integrity standards in the digital learning era in higher education settings.

Keywords Academic integrity, Online examinations, Automated exam proctoring, Remote proctoring, Educational technology, E-assessment

1 Introduction

Conventional examination systems have long witnessed the challenge of maintaining academic integrity. Traditional methods employed in conducting equitable examinations necessitate significant preparation. Students use pen and paper to take a test or



© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

exam, while invigilators are present to ensure the credibility of the examination process. The COVID-19 pandemic necessitated a transition from traditional in-person examinations to digital platforms [1]. Traditional examination monitoring depends on physical invigilation and handwritten responses, wherein proctors supervise student behavior to ensure ethical compliance. However, manually observing student behavior during examinations is time-consuming, and the issue of intellectual dishonesty persists [2].

The rising popularity of online learning over the last 10 years has brought about significant changes in the education sector. Many educational institutions have started using online and blended learning methods, which have made learning more adaptable, customized, and convenient for students [3]. However, as online learning expands, it is necessary to maintain academic integrity and fair assessment practices for students in virtual classrooms [4]. Certain behaviors, such as hiding notes, using technology like mobile phones and calculators, and interfering with other students' work, are considered improper attempts. However, detecting such practices in an online setting is complicated due to the presence of multiple cheating techniques easily accessible to students [5].

Online learning and examinations have had varying effects on students. Some students have shown extraordinary dedication and concentration in their approach to exams [6, 7]. In contrast, others have demonstrated a lack of interest in learning and presenting fair and honest assessments because cheating has become easier for them in the online mode [8]. Apart from this, the prevalence of online exams has introduced psychological stressors that may contribute to dishonest behavior. Students may experience test anxiety, fear of failure, or peer pressure, leading to unethical decisions [9]. A study by Marchant et al. emphasized that academic dishonesty often stems from emotional and psychological stress rather than purely opportunistic motives. [10]. These psychological drivers highlight the need for contextual and supportive monitoring frameworks rather than purely punitive systems.

To overcome these challenges, automated technologies have replaced both manual and online examination systems. Recent advances in Artificial Intelligence (AI) and sensor-based data analytics have propelled a shift from manual proctoring to fully automated systems capable of real-time monitoring, behavioral analysis, and predictive flagging of anomalies [11, 12]. The emergence of automated online proctoring software has systematically enhanced the ability to securely verify the identity of students and monitor their movements during exams. These systems are engineered to accommodate the varying technical and educational requirements of students to ensure that they are viable academically and financially [13].

AI plays a pivotal role in the development of intelligent automated systems designed to detect and deter academic dishonesty during examinations. Several studies confirm that AI's role in academic surveillance systems has become increasingly central due to its ability to scale monitoring, reduce human bias, and adapt to diverse behaviors [14, 15]. These systems, referred to as AIPS, utilize ML and DL techniques based on their specific needs and requirements. Such AI-based systems represent a contemporary and technologically advanced approach to effectively monitor students during online examinations, addressing the limitations of traditional and manual invigilation methods. AIPS ensures system integrity by looking at four main features: (1) authentication, (2) browsing tolerance, (3) remote authorization and control, and (4) generating reports.

The authentication features embedded in it make sure that only registered students can log in and attempt the test. Using cameras for face verification is a popular authentication method. By increasing browsing tolerance, these systems can impose restrictions on students' ability to use their computers and access the internet other than during the tests themselves. The feature of remote authorization and control allows the examiner to initiate, stop, and resume online tests. [16–18]. It also allows them to identify and look into any suspicious activity of students during live proctoring. The last feature, which is report generation, compiles summaries of a student's actions during or after the test by analyzing the recordings of the exam. These automated cheating detection systems utilize front and rear cameras, along with microphones, which are already present on the students' devices, to monitor their actions [19–21].

ML and DL algorithms contribute to the enhancement of the overall functionality of AIPS. ML techniques analyze past exam data, which helps identify student behavior patterns. Apart from past exam data, Many ML techniques are used to analyze the student's behaviors, like the recognition of facial features to detect the movement of their head pose, lips, and eye movements to analyze their activities [22]. It enables the system to detect unusual movements or predict the chances of cheating based on deviations from established behavior patterns. Meanwhile, DL uses Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), Long Short Term Memory (LSTM), and other techniques to analyze the audio, imagery, and video data during exams. These networks can process and interpret complex visual cues such as eye movements, facial expressions, and other forms of body language that may indicate dishonest activities [23–25]. Advanced deep-learning techniques can detect even the most minor forms of cheating. By thoroughly examining sequences of images, these systems can identify any irregularities within a student's surroundings that may indicate the use of unauthorized materials or devices. RNNs are capable of monitoring changes over time, which allows for a detailed analysis of the student's behavior during the exam [26]. AI-driven functionalities automate the detection process, which reduces workforce needs for monitoring. These features allow the system to be more scalable and cost-effective. While the advanced capabilities of AI-driven proctoring systems promote their adoption, they concurrently raise critical concerns regarding user privacy and data security. AIPS is designed to access users' audio and video inputs, and some can even scan for other running programs and restrict user actions. Although these measures are intended to ensure a secure and fair test-taking environment, they may be perceived as violating privacy. So, it is essential to implement measures that assure users of their safety and privacy during test-taking. Various companies set frameworks to ensure privacy standards, and regulatory bodies like the European Union (EU) also release guidelines to regulate data access and storage. The introduction of biometric verification for exams in newer systems has made data security even more crucial, not just during the examination but also for the sensitive data stored and transmitted during the process [27]. This research paper provides an in-depth analysis of AIPS, which uses advanced ML and DL techniques and integrates IoT and biometric systems to detect cheating behaviors among students accurately.

The various components and models presented in Fig. 1 play distinct roles in AI-based proctoring systems. To offer a clearer understanding, Table 1 presents a comparative

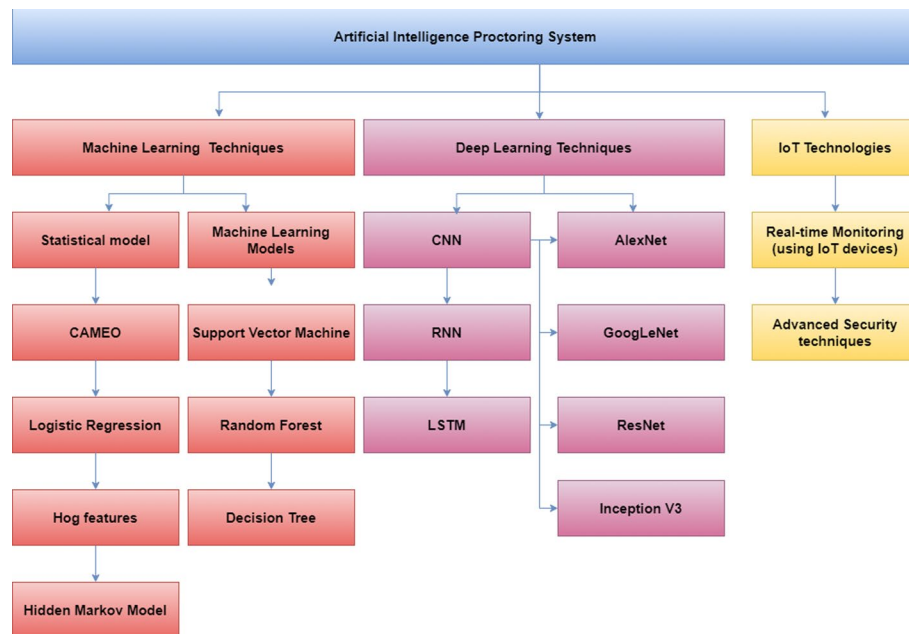


Fig. 1 Classification of artificial intelligence techniques employed in AIPS development

summary of these modules, detailing their primary advantages and potential limitations in the context of automated exam monitoring.

1.1 Research motivation and key contributions

While significant literature exists about AI-based proctoring systems, studies often discuss components of the system in their isolated contexts, but often do not provide a comprehensive, holistic, integrated synthesis. This fragmentation underscores the need for a systematic review that consolidates findings across different studies, highlighting trends, challenges, and advancements in this rapidly evolving field.

Existing reviews [14, 28–31] have approached this issue from limited and point to a gap of studies on integrated/multiple technology assessment as a whole perspectives that do not reflect the broader developments in APS. This paper addresses that gap by presenting a comprehensive examination of the state of AI driven proctoring, considering not only the technologies in use, but also the effectiveness of proctoring systems and the ethical implications of their usage. The scope of the literature considered in this review is confined to higher education settings, as these environments present distinct assessment structures, academic integrity challenges, and regulatory considerations relevant to AI-based proctoring. Through analysis of datasets, architectures, and evaluation clusters and metrics across studies, the review adds clarity to a field where definitions, methods, and reported results are often vague and ambiguous.

Our study aims to bridge this gap by presenting an in-depth analysis of the current state of Automated Proctoring System (APS), including the technologies employed, the effectiveness, and the ethical considerations. Moreover, by examining various datasets and evaluation metrics, this review will elucidate the performance of different APS implementations, offering clarity in an area where confusion often reigns.

Although previous collections of studies situate themselves within studies that examine specific aspects of AI-based proctoring systems, like ML algorithms, authentication

Table 1 Comparative Analysis of the Advantages and Limitations of AI, ML, and IoT Techniques in APS

Model/Technique	Advantages	Limitations
Support Vector Machine (SVM)	Effective for binary classification; works well in high-dimensional spaces	Sensitive to kernel choice and scaling; not ideal for large datasets
Random Forest (RF)	High accuracy; reduces overfitting through ensemble voting	Less interpretable; slower with large numbers of trees
Decision Tree (DT)	Easy to visualize and interpret; fast for small data	Overfitting risk; unstable with small data changes
Logistic Regression (LR)	Interpretable; works well for linearly separable data	Limited to linear relationships; low accuracy on complex data
CAMEO	Useful for detecting timing-based cheating patterns in MOOCs	May produce false positives; assumes user behavior patterns
Statistical Models	Fast, lightweight; suitable for preliminary filtering	Limited adaptability; cannot handle complex input features
Hog Features	Good for shape-based detection; lightweight	Limited robustness to variations in lighting and pose
Hidden Markov Model (HMM)	Useful for temporal behavior modeling	Assumes Markov property; not suitable for high-dimensional input
CNN	Strong at image recognition (e.g., eye/head movement)	Requires large datasets; compute-intensive
RNN	Suitable for sequential/time-series data	Vanishing gradient issue; less effective for long-term dependencies
LSTM	Handles long-term dependencies in behavior sequences	High training time; large memory consumption
AlexNet	Benchmark deep CNN; good for small-to-mid-scale image tasks	Outdated vs newer CNNs; limited depth
GoogLeNet	Inception modules improve accuracy while reducing parameters	Complex architecture; harder to modify
ResNet	Excellent for very deep networks; residual connections solve vanishing gradients	High computational cost
Inception V3	Efficient deep CNN with optimized layers; good for proctoring videos	Complex design; longer training time
IoT Monitoring	Enables real-time, multimodal cheating detection using sensors	Privacy concerns; environment-dependent accuracy
Advanced Security Techniques (with IoT)	Enhances authenticity and data integrity using biometric/fingerprint	Costly hardware; user resistance due to invasiveness

techniques, or evaluation metrics, none provide a complete, integrated synthesis examining the last decade of literature (Table 2). Presently, this study aims to fill that gap through a systematic review of the literature in peer-reviewed articles from 2014–2024, and exploring ML, DL, IoT, and hybrid methods. The last 10 years were selected because APS research has significantly advanced during this period due to breakthroughs in AI, ML, DL, computer vision, biometric authentication, and scalable cloud-based assessment platforms, alongside widespread adoption during the COVID-19–driven shift to remote education. Additionally, this review draws on literature pertaining to AI-based proctoring within higher education institutions. Unlike previous reviews, a systematic review synthesizes technology, methodology, and ethics; assesses the integration of hardware and performance metrics; and uses a conceptual model to explain an integrated understanding. This integrative approach creates a portrait of trends, highlights the perception of challenges, and provides actionable insights for researchers and practitioners of AI-driven proctoring systems.

As automated proctoring continues to gain traction in response to the growing demand for innovative assessment methods, there is an urgent need for a study that explores its progress and the associated challenges comprehensively. This paper serves

Table 2 Comparative analysis of existing reviews on AI-based proctoring systems

References	Focus area	Scope and coverage	Limitations identified	Relevance to current review
[14]	AI in academic proctoring	Comprehensive literature synthesis of AI strategies in academic student proctoring	Limited model evaluation of datasets and does not address technical analysis across a system architectural level	Describes a clear gap of having deeper synthesis of algorithm and datasets
[28]	AI-based proctoring systems	Articulates the historical advancement of AI, current implementation approaches, and potential future trends	Limited dataset review does not include the consideration of multi-modal assessment fusion or robustness	Provides a solid background but lacks depth of technical analysis
[29]	Digital proctoring in higher education	Emphasizes institution-based considerations, student perceptions, and managerial implications	Generates no analysis of technical algorithms, datasets, or technical performance metrics	Provides background information and context, does not include technical synthesis
[30]	Accuracy of auto-mated AI proctoring	Considers accuracy-focused research in online student exam assessments	Extremely limited in scope; employs accuracy arguments and associations without technical structures	Reiterates need for multi-factorial technical assessments and ethical considerations
[31]	ML-based facial fraud detection	Analyzes facial dataset ML approaches for detecting student cheating	Extremely limited on modality; does not address ethics, system architecture, or multi-modal coverage	Points to a gap of studies on integrated/multiple technology assessment as a whole

as an entry point for researchers delving into this critical area, providing a kick-start for future investigations. By systematically cataloging and analyzing the literature from the past one decade, we aim to present a clear picture of how APS has developed and the ways in which it can be further enhanced to meet the ever-changing demands of educational assessment.

The key contributions of this study are highlighted below:

1. Provides a comprehensive review of 80 academic publications (2014–2024) that apply ML, DL, IoT, and hybrid models in APS.
2. Identifies the strengths and limitations of various AI models such as CNN, RNN, SVM, and RF in the context of detecting cheating behavior.
3. Classifies proctoring systems based on the type of AI techniques used ML, DL, IoT, and hybrid integration approaches.
4. Proposes a conceptual multi-layered framework integrating biometric verification, visual/audio analysis, and environmental monitoring to improve academic integrity.
5. Highlights ethical concerns related to student privacy and outlines compliance considerations with data protection standards such as the General Data Protection Regulation (GDPR).

The paper is structured as follows: Section 2 details the PRISMA methodology employed for the literature selection. Section 3 provides an extensive review of various AI-driven proctoring approaches, encompassing ML, DL, IoT-based systems, hybrid models, and alternative AI frameworks. Section 4 summarizes the key findings derived from the literature, while Section 5 examines the future scope and the inherent challenges within the field. Section 6 presents a critical discussion of the current landscape, and finally, Section 7 concludes the study with actionable insights and recommendations.

2 Methodology

This review adopts a rigorous and systematic methodology to identify, select, analyze, and synthesize the literature pertaining to AIPS in higher education. To ensure academic rigor and transparency, this study adopts a formal systematic review methodology, following established protocols including Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA). The primary objective of this review is to synthesize peer-reviewed research on AIPS used in higher education between 2014 and 2024, particularly focusing on ML, DL, and IoT based solutions.

2.1 Research questions

The review is guided by the following core research questions: RQ1: What are the main AI techniques and system architectures used in APS for academic integrity? RQ2: How effective are these systems based on technical performance metrics such as accuracy, precision, recall, and AUC? RQ3: What are the ethical, practical, and deployment-related challenges associated with AIPS? RQ4: What future directions, open challenges, and research gaps exist in this domain?

2.2 Systematic review protocol

The systematic review follows a multi-phase methodology aligned with PRISMA standards:

1. *Identification*: Retrieval of studies using structured queries from databases.
2. *Screening*: Removal of duplicates, followed by abstract and title-based filtering.
3. *Eligibility*: Full-text review based on inclusion/exclusion criteria.
4. *Inclusion*: Final selection of studies meeting all methodological standards.

2.3 Research design

This study adopts a systematic review methodology following PRISMA guidelines, incorporating qualitative and quantitative synthesis of the selected studies. The approach enables the analysis of both experimental results and conceptual frameworks across various AI-based proctoring systems. The objective is to explore trends, identify gaps, and assess the effectiveness and ethical implications of existing systems.

2.4 Search strategy

To ensure a comprehensive and representative collection of scholarly works on AIPS, a structured and methodologically rigorous search strategy was employed. The literature search spanned across five major academic databases renowned for their high-quality, peer-reviewed content: IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect, and Google Scholar. These platforms were selected to cover a broad spectrum of engineering, computer science, and educational technology publications, thereby maximizing the diversity and relevance of the retrieved literature. In this systematic review, search outcomes from Google Scholar were used to serve as an adjunct resource to the primary database search to identify peer-reviewed conference papers and journal preprints that did not appear in other databases and not to report grey literature, e.g., unpublished reports and white papers. This distinction was made for purpose of clarity on the inclusion criterion which required peer-reviewed research; however, still broaden

the coverage to newer and emerging research publications. The query was narrowed to articles published from January 2014 to March 2024 to examine the before and after developments in digital proctoring due to the COVID-19 pandemic. This timeframe was chosen to comprehensively cover the emergence, acceleration, and refinement of digital proctoring solutions in both pre-pandemic experimentation and post-pandemic adoption phases. The search was conducted using Boolean combinations of the following core keywords: “AI proctoring”, “automated exam monitoring”, “machine learning cheating detection”, “deep learning proctoring system”, “IoT-based invigilation”, “online exam integrity”, “virtual invigilation”, “remote proctoring”, “academic dishonesty detection”, and “intelligent surveillance system”(“AI proctoring” OR “automated exam monitoring” OR “machine learning cheating detection” OR “deep learning proctoring system” OR “IoT-based invigilation” OR “online exam integrity” OR “virtual invigilation” OR “remote proctoring” OR “academic dishonesty detection” OR “intelligent surveillance system”). These terms were selected based on domain relevance, recurring vocabulary in existing literature, and indexation consistency across major databases.

The primary search yielded a total of 348 records. The vast amount of studies highlights the strict inclusion process by concentrating on peer-reviewed empirical studies and removing duplicates to only include studies of methodological rigor. Following screening of titles and abstracts and full text eligibility review, 80 studies met all inclusion criteria. These are the studies included in the final synthesis and mentioned in Table 3.

2.5 Inclusion and exclusion criteria

To ensure methodological rigor relevance, transparency, and reproducibility, following inclusion and exclusion criteria was applied in screening and retrieval process:

2.5.1 Inclusion criteria

1. Peer-reviewed journal or conference publications.
2. Published between January 2014 and March 2024, the evolution of the automated proctoring system.
3. Written in the English language.
4. Focused on AI-based proctoring in higher education.
5. One or more computational approaches must be proposed, executed, or assessed using ML, DL, IoT, or hybrid techniques.

2.5.2 Exclusion criteria

1. Editorials, opinion pieces, and non-empirical studies without system implementation.
2. Papers not directly addressing exam proctoring or academic integrity or assessment-related misconduct.
3. Any paper other than English language.
4. Duplicate records or papers lacking a technical/systemic description.

Table 3 Year-wise distribution of selected articles (n = 80)

Year	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Count	1	3	5	7	2	3	11	10	17	18	3

Studies that did not provide a technical/systematic description and therefore were operationalized as follows: A paper was excluded if it did not describe at least two of the following essential methodological components needed for reproducibility:

1. System or model architecture such as algorithms, modules, and workflows.
2. Dataset characteristics (where the data is sourced from, sample size, data preparation steps, or data modality).
3. Evaluation process (operationalization of experiment, metrics, or validation process).

Studies that did not include these could not be reasonably compared to other implementations and were therefore excluded from analysis.

Exclusion summary

A total of 348 records were identified. After omitting 68 duplicate articles, a total of 280 titles and abstracts were screened by the research team. A total of 143 full-text articles were reviewed for eligibility and 80 studies were eligible for inclusion in the final review.

The 63 articles excluded were classified as follows:

1. 23 Studies did not report empirical research, including editorials, opinion pieces, and conceptual papers
2. 13 Records referred to studies unrelated to exam proctoring or studies examining academic integrity
3. 27 Studies failed the reproducibility criterion of the above statistical operational definition

The processes above ensured that only methodologically rigorous and technically interpretable studies were included for review.

2.6 Screening and selection process

The screening and selection procedure proceeded in a structured, multi-stage workflow. The comprehensive and exhaustive search in the specified academic databases generated a total of 348 records. Therefore, the removal of 68 duplicates left 280 unchanged articles for screening. Screening of records was completed by two independent raters with computer science and AI-based educational technology expertise. Both reviewers screened titles and abstracts of all 280 records for preliminary relevance to the review process. Any disagreements were resolved through discussion. This process ensured that screening judgments were transparent and unbiased. The title and abstract screening phase lasted around two weeks and the full-text assessment of the 143 papers took four weeks. Overall, the entire screening process lasted around six weeks, including conflict resolution and the documentation of decisions at each phase. Thereafter, in the full-text phase, the reviewers assessed each publication against the previously established inclusion and exclusion criteria, ultimately arriving at a final number of 80 articles that met all empirical rigor, technical detail, and theme relevance study criteria. The full selection process is illustrated in the PRISMA diagram in Fig. 2. To facilitate thematic synthesis and cross-study comparison across papers included, each paper was analyzed manually using a structured framework. The reviewers analyzed and categorized the information within each paper along four key areas of focus. Primary AI techniques: ML, DL, IoT, or Hybrid. Proctoring content: including distinctions between real-time or post-hoc

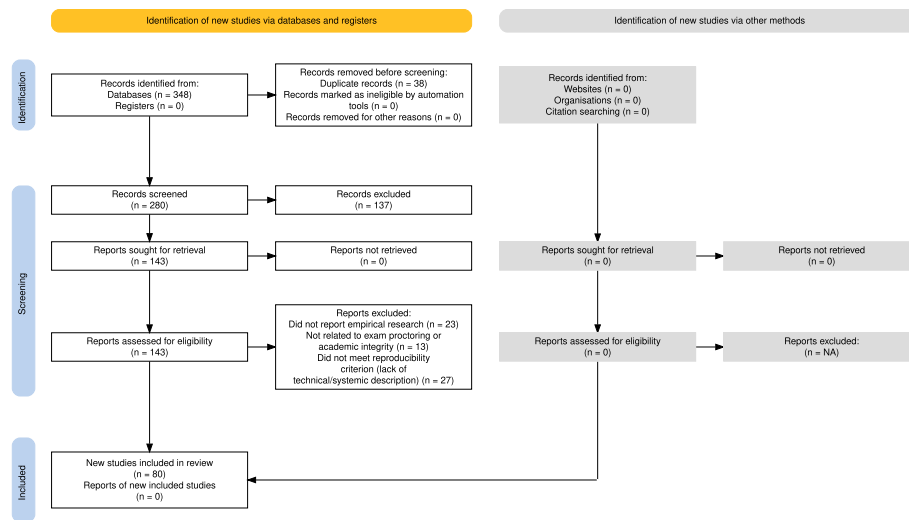


Fig. 2 PRISMA flow diagram showing the article selection process for this review

Table 4 Metadata elements extracted from selected studies and their purpose

Metadata element	Description/purpose
Title, Authors, Year	To monitor chronological trends and identify contributions over a time period
AI Technique (e.g., CNN, SVM, RF, LSTM)	To identify methodological diversity and preferences in the literature
Application Scope	To capture functional areas including face detection, gaze tracking, voice and sound assessment, keystroke logging, and multi-modal behavior recognition
Dataset Type	To evaluate reproducibility and robustness (open-access, custom, or simulated datasets)
Performance Outcomes	To facilitate comparative benchmarking using metrics, e.g., accuracy, precision, and recall
Ethical/Privacy Observations	To analyze socio-technical impact, including GDPR compliance, or biometric processing

and online or offline exams. Performance metrics (when reported), such as accuracy, precision, recall, and AUC. Integration, including any software or hardware such as a webcam, microphone, gaze tracker, keystroke sensor, or biometric. A hybrid synthesis methodology was implemented, comprising descriptive synthesis for quantitative performance patterns and thematic mapping to identify design trends, application domains, and common technical challenges. This combination allowed us to productively interpret literature and take an approach that went beyond a narrative review and offered a more structured, and evidence-based synthesis of the current state of knowledge regarding AI-driven proctoring systems.

2.7 Data extraction and synthesis

To ensure a thorough and meaningful synthesis of the selected literature, a structured data extraction protocol was employed. For each of the 80 selected studies, a comprehensive set of metadata elements was meticulously recorded to facilitate both quantitative and qualitative analysis. These elements and their purpose are summarized in Table 4.

The extracted data was systematically synthesized using a combination of comparative tables, feature-specific figures, and model taxonomy charts, which provided

multi-layered insights into the structure, capabilities, and limitations of AIPS implementations. This dual approach of quantitative evaluation of model performance and thematic mapping of application domains enabled a holistic exploration of current research trends, design patterns, and technical challenges. The result is a rich, multi-dimensional understanding of the state of AI-based proctoring systems, supporting the identification of research gaps and informing future innovations in the domain of academic integrity enforcement.

2.8 Quality assessment

In order to maintain the review's methodological rigor and reliability, all 80 studies were systematically reviewed and assessed for their quality. The criteria for assessment included: (i) technical robustness of the AI approach, (ii) clarity around the design of the system architecture, (iii) clarity on the dataset (e.g., availability for review, transparency and reproducibility), (iv) ethics including privacy and consent in the data or AI methodologies, and (v) relevance to the assessment context in higher education. For studies meeting these criteria, they were included for analysis, while studies lacking clarity on technical robustness or ethics were either removed from comparative tables or kept for narrative purposes. Table 5 summarizes the quality assessment of the studies included in these reports, which outline the validity of the synthesis.

For methodological consistency, each of the 80 selected studies was evaluated against each of the seven criteria in Table 5 with a checklist strategy, which included yes/no answers and qualitative notes. This process facilitated a transparent comparison of technical competency, dataset description, ethical clarity, and methodological completeness. The challenges summarized in Table 6 directly corresponded to gaps that were experienced during this evaluation, linking the quality evaluation to challenges within the literature as presented.

Table 5 Quality assessment of selected studies on AI-based proctoring systems

Quality criterion	Framed questions	Approach	Notes
Technical Soundness	Are AI, ML or DL techniques appropriate and validated?	Check CNN, SVM, RF, LSTM, or hybrid methods that are aligned with standard metrics	Highlight studies with rigorous evaluation metrics
System Architecture Clarity	Are the workflows, modules, and interactions clearly elaborated?	Evaluate architecture, diagrams, explanations, and reproducibility potential	Papers lacking clarity are omitted
Relevance to Higher Education	Does the paper target online or hybrid exam proctoring?	Include papers focused on educational assessment settings	Exclude unrelated AI applications
Dataset Transparency & Reproducibility	Are dataset details reported (size, source, processing)?	Assess whether datasets are open-access, simulated, or custom	Transparency improves replicability
Performance Evaluation	Does it mention metrics (accuracy, precision, recall, AUC)?	Compare quantitative outcomes	Missing metrics are omitted
Ethical & Privacy Considerations	Does the paper discuss privacy, consent, bias, or compliance (GDPR, PDP)?	Include if clear policy or ethical discussions exist	Important for socio-technical responsibility
Publication Quality	Is it peer-reviewed? Consider journal impact factor or conference reputation	Check IEEE, ACM, Springer, Scopus-indexed, etc	Supports credibility and reliability

Table 6 Summary of challenges identified across selected AI-based proctoring system studies

Challenging category	Challenges description	Frequency/observation
Dataset Limitations	Small sample size, not open-access dataset, inconsistent modalities.	35% of the studies did not fully disclose their dataset.
Technical Limitations	Limited validation of AI models, no benchmarking across models.	40% of the studies only evaluated a single model.
Reproducibility Issues	Insufficient information on architecture or experimental workflow.	30% of the studies could not be fully reproduced.
Ethical or Privacy Concerns	Lack of privacy compliance, missing consent, insufficient handling of bias.	25% of studies explicitly mentioned ethical issues.
Performance Evaluation	Inconsistent or missing reporting of metrics such as area under the curve or recall.	20% of studies did not report standard metrics.

2.9 Challenges and limitations in literature

Through the quality assessment and thorough analysis, we determined some challenges and limitation in the existing selected studies mentioned in Table 6.

2.10 Ethical and practical considerations

In addition to evaluating the technical sophistication and empirical performance of AIPS, this review places significant emphasis on the ethical and practical dimensions of deploying such technologies in educational settings. As AI-driven monitoring tools increasingly access sensitive personal data, such as facial imagery, voice patterns, gaze movements, and keystroke dynamics, data privacy and user consent emerge as critical concerns. Many reviewed systems engage in continuous audiovisual surveillance, necessitating clear communication of data collection policies, explicit user consent mechanisms, and secure handling of biometric and behavioral data. The presence or absence of such ethical safeguards was carefully noted during the review process. Furthermore, the issue of algorithmic fairness and bias was assessed across systems that rely on computer vision and pattern recognition. Several studies acknowledged that model accuracy can vary across demographics, lighting conditions, and hardware setups, potentially disadvantaging certain student populations. The review also evaluated how well systems address equity and inclusivity, especially in geographically diverse or resource-constrained environments. Regulatory compliance was another cornerstone of ethical assessment, many systems claimed alignment with global data protection frameworks such as the GDPR in Europe and the Personal Data Protection (PDP) Bill in India. Where applicable, institutional policy adherence and ethical review board approvals were also considered as indicators of responsible development and deployment.

3 Literature review

3.1 Key issues and solutions in automated proctoring systems

APS have grown in popularity due to the rise of online education, but they also face several critical challenges that impact their effectiveness and acceptance. One of the primary challenges in AI-driven surveillance systems is the prevalence of false positives and false negatives. Systems may incorrectly flag normal student behavior, such as looking away from the screen or adjusting their posture, as suspicious activity, while simultaneously missing subtle instances of cheating. This affects the reliability of the system and undermines student trust [32, 33]. While existing studies offer valuable insights into specific applications of ML, DL, and IoT in proctoring systems, a critical comparison reveals varying levels of generalizability, dataset robustness, and real-time efficacy. For instance,

while CNNs have high visual accuracy, they demand extensive training data, which may not generalize across demographic settings. RNN-based models perform better in temporal analysis but are prone to vanishing gradients and limited scalability. Moreover, many studies rely on controlled or simulated datasets, limiting external validity. Comparative evaluations often overlook trade-offs between precision and invasiveness, which are central to ethical deployment. Another challenge is the limited generalizability of current AI models. Most systems are trained on narrow datasets, often lacking demographic and environmental diversity. As a result, performance significantly drops when deployed in real-world scenarios with varied lighting conditions, cultural behaviors, or physical settings [34]. Moreover, privacy concerns are a consistent barrier to user adoption. Continuous video and audio surveillance, combined with invasive data storage practices, raise ethical questions and compliance issues with frameworks like GDPR and India's PDP Bill [35]. Technical constraints such as device dependency and unstable internet connectivity further exacerbate the inequity in online assessments. Students in rural or low-bandwidth areas are disproportionately affected, facing exam interruptions or system malfunctions [36]. Additionally, current proctoring solutions often lack contextual awareness; they may detect movement or sounds but are unable to interpret their intent e.g., distinguishing a stretch from a signaling gesture [37]. To address these challenges, recent studies have proposed integrating hybrid AI models that combine DL with rule-based decision logic to reduce false positives. For instance, CNNs enhanced with behavior-tracking algorithms and gaze estimation modules have shown improved accuracy in detecting anomalies [38]. Efforts are also underway to diversify datasets by including videos and behaviors across geographies, enhancing model robustness [39]. In terms of privacy, implementing privacy-preserving architectures such as edge processing, anonymization techniques, and secure on-device data handling has been recommended [40]. To tackle infrastructural limitations, lightweight AI models optimized for low-resource environments are being explored, using tools like TensorFlow Lite and Jetson Nano for local processing. Finally, multimodal approaches combining facial recognition, voice analysis, keystroke dynamics, and system activity offer a more nuanced understanding of user behavior and reduce reliance on any single input stream [41]. Despite significant advances in APS using ML, DL, and IoT technologies, a clear, unified framework that balances detection accuracy, privacy preservation, and context-awareness remains lacking. Many existing models are limited by dataset biases, generalization issues across diverse environments, and ethical concerns around intrusive surveillance. Moreover, there is insufficient integration of multimodal behavioral inputs, such as keystroke dynamics, gaze tracking, and environmental monitoring, in a cost-effective and scalable way. These limitations indicate a critical need for developing hybrid systems that are not only technically robust but also ethically and practically implementable in real-world educational settings.

3.2 Machine learning-driven automated proctoring system

The integrity of virtual examinations is ensured by the use of ML techniques in automated proctoring procedures. These systems examine a variety of behavioral cues, such as keystrokes, browser activity, eye movements, and facial expressions, to identify any indications of cheating. APS may detect suspicious trends by continually monitoring these behavioral indicators. They can then warn instructors or invigilators in real-time,

offline, or via recorded online tests, so they can respond promptly and appropriately. Reliable proctoring solutions are actively becoming necessary as digital tests and remote learning grow more mainstream. ML algorithms possess the capacity for continuous improvement, enabling the detection of emerging patterns of academic dishonesty. The prior research on ML approaches and their role in automated proctoring system development is examined in this section.

R.S et al. [42] use a heuristic-based proctoring system to analyze cheating behavior, which involves the usage of webcams and microphone devices. Cheating behaviors were monitored using techniques such as facial detection, detection of user absence, presence of multiple individuals, voice recognition, gesture analysis, and active window tracking. The system performs quite well, with a 0.08 false-positive rate and a 0.13 true-negative rate.

Li et al. [43] introduce a Massive Open Online Proctoring (MOOP) framework designed to efficiently proctor online exams. The main components of the methodological approach are the Automatic Cheating Detector (ACD), Peer Cheating Detector (PCD), and Final Review Committee (FRC). The ACD uses various sensor technologies, like webcams, EEG sensors, and gaze trackers, to automatically detect cheating behaviors. Peer students evaluate the cases of cheating in ambiguous cases that ACD has flagged for review by PCD. Lastly, the FRC reviews these cases to decide on any punitive actions. The experiments show that the webcam-only version of ACD achieved a recall of 78.6%, precision of 84.6%, and accuracy of 83.3%. On the other hand, the multi-modal ACD, which includes more sophisticated sensors, displayed improvements in recall but at the expense of precision and overall accuracy. The experiment involved an exam scenario with computer science students where multiple-choice questions were administered through a platform that could synchronize sensor data (like gaze and EEG) with the test responses. The hardware setup included Microsoft Kinect cameras for capturing video, the Mirametrix S2 eye-tracker for gaze tracking, and the NeuroSky MindSet for EEG monitoring. These devices played a critical role in collecting multimodal data used for training and validating the cheating detection algorithms.

Soma et al. [44] focus on addressing the problem of fraud detection in offline examination hall tickets using an automated system. To detect fraud in examination hall tickets, the methodology uses an Artificial Neural Network (ANN) classifier to authenticate candidate identities by analyzing textual (name and seat number) and graphical (photograph) elements on the ticket. The process involves several key phases, such as prepossession, segmentation, face detection, feature extraction, and ANN classification. The ANN classifier is trained to differentiate between original and fraudulent hall tickets based on a dataset comprising images from two different institutions. The proposed system was tested on a significant dataset, demonstrating an average accuracy of 94.12% in identifying fraudulent tickets. The study used a dataset of hall ticket images from two different institutes, which included various features such as candidate names, seat numbers, and photographs. This dataset was essential for training and testing the ANN classifier.

The problem of detecting cheating via whispering has not received enough attention, according to Abdullah et al. [45] especially when visually challenged and deaf people are involved as invigilators. The authors' technique comprises four primary components: energy estimation, spectrum estimation, time length estimation, and silence removal.

The features extracted from the whispered signals include energy, Root Mean Square (RMS), time duration, and spectral features. The Z-Score is utilized for threshold evaluation. The research collected 50 samples of different types of sounds to use as datasets, such as sounds made while speaking, combined sounds, hushed speaking, sneezing, coughing, walking sounds, turning pages, and clicking a pen. Each sound was recorded for 10 s and had a sampling frequency of 11,025 Hz. The proposed method achieved satisfactory results with a False Acceptance Rate (FAR) of 1 percent and a False Rejection Rate (FRR) of 3 percent in analyzing 100 sample signals.

AlSallal et al. [46] highlight that the existing plagiarism detection methods have limitations due to their rigid assumptions, such as assuming that the plagiarists copy and paste directly from other authors, and can be easily tricked by rewording or summarizing text while retaining the actual meaning. The authors propose an integrated approach using Bag of Words (BOW), Latent Semantic Analysis (LSA), Stylometry, and SVM. BOW breaks a document into unique words and calculates the frequency of each word, while LSA provides a quantitative representation of a semantic domain. Stylometry aims to characterize an author's writing style based on features like Most Common Words (MCW) and Content Words (CW). Lastly, the SVM is used as a classifier to detect these features. The Corpus of English Novels (CEN) dataset is used for this study. The dataset includes 292 novels by 25 authors, with a total of 25 million words, and each author's novels were analyzed separately.

Cote et al. [47] propose an innovative approach of video summarizing for remote invigilation of online exams by modeling normal and abnormal student behavior patterns. They did this by simulating normal and abnormal student behavior patterns. The absence of automated methods that can effectively detect and summarise possibly anomalous behavior during online tests is the main research gap that is being addressed in the study. Using a two-state Hidden Markov Model (HMM), the suggested method offers a novel solution to this problem by predicting behavior among students based on head position cues. The study's approach consists of a few crucial components, such as MATLAB, which is used for behavior modeling, followed by Python and OpenCV for head pose estimation.

Staubitz et al. [48] compare and contrast the most widely used online proctoring techniques and develop and deploy a novel automated proctoring strategy based on face recognition technology. Webcams and standard computers are used to perform this study and arrive at conclusions. The research employs a statistical model to match participant faces against previously submitted images, with a particular focus on automating identity verification without ongoing human supervision. The paper provides results from an experiment using their automated proctoring system, which shows the technology would effectively increase credibility in online certificates by strengthening the connection between the evaluation procedure and the participant's identification. The results suggest that, while the automated system can not guarantee absolute security against fraud, it significantly enhances the reliability of the certification process compared to unproctored assessments. The authors used information from the Hasso Plattner Institute's MOOC platform, which included responses from many user surveys about the value of certificates and their opinions of online proctoring, to assess the efficacy and acceptability of this methodology.

The risk of cheating by using numerous accounts is a significant problem with Massive Open Online Courses (MOOCs), according to recent research by Northcutt [49]. To deal with this issue, the research presents the Camera Assisted Meeting Event Observer (CAMEO) detection method, which looks at user interaction data to identify instances of cheating. The technique used by the algorithm is filter-based and focused on the time differences between retrieving answers and sending accurate replies. The system is made more resilient and dependable by using statistical models and Bayesian criteria to predict the probability of cheating behavior. According to the data analyzed by this study, over 1.9 million students took part in 115 MOOCs that were provided by MIT and Harvard.

Chuang et al. [50] focus on filling the gap in the role of time delay and variations in a student's head pose relative to the computer screen in predicting cheating behaviors for online examinations. The study involved 42 undergraduate students from a public university in the United States. Participants underwent a series of testing phases, which included learning sessions, online testing in typical and cheating-inducing environments, and interviews. Various features related to Visual Focus of Attention (VFOA), including head pose variations, were extracted from webcam videos. The Fast Correlation-Based Filter (FCBF) method was employed for feature selection, while LR and hierarchical logistic regression were used to analyze the data and develop predictive models. The study found that both time delay and head pose variations relative to the computer screen were significant predictors of cheating behaviors. The final LR model, incorporating these predictors, achieved an average accuracy of 75.6% in detecting cheating incidents during online exams. The primary device used for data collection was the webcam, which captured participants' head movements and behaviors during online exams.

Ruipérez et al. [51] have identified the need for the implementation of an ML technique to accurately detect cheating behavior, especially in MOOCs where IP address detection is either unreliable or impractical. The authors employed the RF algorithm to detect instances of academic dishonesty. This involved a meticulous process of selecting and analyzing student performance-related attributes to train the RF model. The model's effectiveness was evaluated using various metrics such as accuracy, sensitivity, specificity, and the Area Under the Curve (AUC), and it was trained with labeled data from both CAMEO and non-CAMEO submissions. The dataset encompassed attributes such as student performance, video-watching behavior, problem type, submission time, and required attempts. The RF model exhibited robust performance in identifying instances of cheating in MOOCs, demonstrating high sensitivity and specificity. Notably, the model excelled at recognizing CAMEO inputs, as indicated by its nearly flawless AUC score of close to 1.

Jalali et al. [20] proposed a methodology to detect cheating behavior using image processing. The proposed method uses capturing images of students during exams, which are then analyzed and compared with reference images to detect any instances of cheating behavior. The proposed system detects certain conditions, like students sitting behind a desk with only a computer, consistent lighting, and webcam angles. The authors used a custom dataset consisting of 50 reference images of each student in a normal state, along with 10 images of each student in both normal and cheating states during the exam. According to the research findings, the suggested technique can flawlessly identify vacant seats created by students leaving their chairs. The detection accuracy rate is recorded as 100 percent. The study further introduces two distinct methodologies for

detecting academic dishonesty. The first method has a higher accuracy rate, whereas the second method is more time-efficient.

Atoum et al. [52] observe that current methods involving human proctoring via webcams or testing centers are not scalable and can be quite costly. Therefore, there is a crucial need for an automated system to monitor and detect cheating in online exams to ensure the credibility of online education. The proposed system combines hardware and software components to detect cheating behaviors. The hardware includes a webcam and a microphone to monitor the visual and acoustic environment. The software comprises six components: user verification, text detection, voice detection, active window detection, gaze estimation, and phone detection. By analyzing audio-visual data in real time, these components work together to identify potential cheating.

A proposed method by Soman et al. [53] uses video analytics to detect suspicious activities during examinations. The methodology incorporates a Histogram of Oriented Gradients (HOG) feature and a K-Nearest Neighbors (k-NN) classifier to analyze and classify behaviors as normal or abnormal, including cheating behaviors such as peeking, passing notes, and other suspicious gestures. The system is adaptable and customized to specific examination hall conditions. It is notably fast at detecting activities, with normal behaviors identified in approximately 0.000,003 s and various abnormal behaviors (peeking, signaling, turning around, and passing paper) detected within about 0.920118 to 1.253,411 s. With a recall rate of up to 94.4% for passing papers and precision of up to 94%, the system demonstrates high accuracy in recognizing these actions. The data set used to train and test the classifier consisted of real-time videos recorded in a classroom setting, featuring both single and multiple rows of students engaging in different activities at a 720x1280 resolution.

Nagoudi et al. [54] focus on detecting cheating behaviors in Arabic text such as synonym substitution, paraphrasing, and text structure manipulation, which are not adequately addressed by existing tools. Two methods were introduced to detect plagiarism, even in cases where it is skillfully disguised. The first method utilizes word embedding, word alignment, and word weighting to assess semantic relationships in texts. The second method employs ML classifiers, such as SVM, DT, and RF, to distinguish between plagiarized and non-plagiarized text segments using lexical, syntactic, and semantic features. Using a specific parameter setting ($k=15$), the word-embedding approach achieved a Plagdet score of 84.47%, with a recall of 87.64% and a precision of 87.93%. The ML approach, using the SVM classifier combined with all features, achieved a plagiarism detection score of 86.7%, with recall at 92.1% and precision at 88.6%. The evaluation was conducted on the EXARA-2015 corpus from the 1st International Competition on Arabic Plagiarism Detection, which contained various types of plagiarism cases in Arabic texts.

AlSallal et al. [55] work on intrinsic plagiarism detection, which refers to the identification of instances of plagiarism without relying on a reference collection. To address this challenge, an integrated approach has been proposed by the authors that combines the statistical properties of common words, Latent Semantic Analysis (LSA), and ML algorithms such as Multilayer Perceptron (MLP). This method generates a model of the author's style by analyzing the usage patterns of the most common words. It aims to detect subtle variations in text that may indicate plagiarism. To evaluate the effectiveness of this approach, the authors utilized the CEN dataset, which consists of 292 novels

written by 25 British and North American authors. The results of the study showed an impressive overall accuracy of 97% in authorship prediction, which is a significant achievement for intrinsic plagiarism detection tasks.

While other scholars work on detecting text plagiarism using ML methods, Qiubo et al. [56] worked on code plagiarism. The detection of code plagiarism in programming assignments is a critical task that has traditionally relied on similarity thresholds. However, this research reveals that the previously used methods have several deficiencies that could lead to high rates of false positives or negatives. To address this issue, the authors of a recent study propose an innovative approach that combines RF and Gradient Boosting Decision Tree models for evaluating code plagiarism. This new methodology dynamically adjusts similarity thresholds and considers additional factors beyond code similarity, such as programming style and student submission behavior, to enhance plagiarism detection accuracy. The proposed model is based on a dataset of over 300,000 code submissions, with a subset of 15,979 cases initially marked as potential plagiarism for in-depth analysis. The results are impressive, with an accuracy rate of 95.9% in distinguishing between plagiarized and non-plagiarized submissions, which is significantly higher than the baseline method. The authors' approach offers a promising solution to the problem of detecting code plagiarism in programming assignments and could be of great value to academic institutions and businesses alike.

Sangalli et al. [57] focus on detecting unethical behaviors in online learning environments, with a specific emphasis on identifying students who cheat by sharing answers or using fake accounts to harvest correct answers. The proposed metrics are based on co-occurring events and interactions within the course and are designed to flag potential cheating. By analyzing pairs of user accounts that solve exercises closely in time and computing metrics for these pairs, the study can distinguish between real student collaborations and those using fake accounts to cheat. The method utilizes K-means clustering, and the research drew on data from the online course "Jugando con Android: aprende a programar tu primera app" offered by the Universidad Autónoma de Madrid. The dataset includes logs of student interactions, exercise submissions, forum posts, and other relevant activities within the course. The methodology achieved an impressive generalization accuracy of over 95% in identifying these types of cheating using an SVM. Detecting plagiarism in student homework, especially programming assignments, is challenging.

Ljubovic et al. [58] has identified a gap in current tools that mostly focus on code similarity. Students using advanced methods to evade detection pose challenges in identifying external plagiarism. The authors introduce a novel methodology that utilizes a cloud-based integrated development environment (IDE) to log student activities while they solve homework. The approach involves creating ultra-fine-grained repositories of code changes that enable the extraction of detailed developer profiles. To detect plagiarism, authors use ML techniques to analyze profiles and identify suspicious behavior. This approach has proven to be more effective than traditional code similarity methods. The dataset allowed the extraction of features related to students' interaction with the IDE, offering a novel perspective on behavior indicative of plagiarism.

Sinha et al. [59] highlight the need for a more robust system that can ensure the integrity and fairness of remote examinations. The authors used a combination of hardware and software technologies, integrating AI and ML to create a secure and efficient exam

environment. The system requires examinees to register and verify their identity using webcams and microphones, and it uses these devices during the exam to track eye and head movements, aiming to detect and deter cheating. The system incorporates robust security features like encryption, secure data handling, and a lock-down of the examination interface to prevent access to unauthorized applications during the test. The hardware requirements include a webcam and microphone, with optional secondary cameras for enhanced monitoring. Although the paper outlines a theoretically sound framework for a remote proctoring system that could promote online examinations by making them more secure and less prone to fraud, it lacks empirical data on the system's effectiveness and user compliance. Additionally, the results concerning the system's accuracy in detecting dishonest behavior or its usability in varied real-world scenarios are not detailed. This lack of empirical evaluation suggests a need for further research to validate the effectiveness and practicality of the proposed system in diverse educational settings.

Hoque et al. [60] propose an automated exam invigilation system that integrates biometric authentication with CCTV surveillance to streamline the traditional exam monitoring process. This system uses fingerprint sensors to ensure that only registered students can enter the examination hall, while 360-degree CCTV cameras and ultra-sensitive microphones monitor the exam environment to prevent and detect any malpractices. The methodology aims to reduce the number of invigilators needed, thereby cutting down the cost and complexity of traditional exam invigilation. The system also effectively authenticates students' identities, which prevents unauthorized exam participation. The implementation of this system shows promising results in enhancing security and integrity by significantly reducing the chances of malpractice. The proposed automated system is expected to be more cost-effective compared to traditional methods, as it requires fewer human resources and less logistical preparation.

Yaqub et al. [61] highlight the need for a system that can detect cheating attempts during exams while preserving the privacy of students. The proposed system deploys image hashing-based anomaly detection to identify excessive face and body movements indicative of cheating, even when the student's face is blurred or masked in video frames. The system's potency is assessed using an in-house dataset of exam-taking videos and public image datasets. Ultimately, it attained an average accuracy of approximately 75% across participants when using masking-based privacy-preserving measures. Additionally, the system demonstrates low computation overhead, with both blurring and masking approaches easily executable on standard PCs.

Li et al. [62] have developed a state-of-the-art visual analytics approach that utilizes webcam video recordings and mouse movement data to detect suspicious activities during online exams. The methodology includes creating a mock online exam to collect the necessary data, implementing a web-based system for data collection, and employing a visual analytics system to analyze and present the data for proctoring purposes. The system displayed impressive results in identifying cheating behaviors, outperforming the baseline method in both accuracy (0.888 vs. 0.606) and efficiency (time taken: 374.375 s vs. 463.938 s). The mock online exam involved 24 participants, generating 48 videos and 286,940 mouse movement records. The system proves to be viable for observing and analyzing students' behavior during exams by making it more efficient and less labor-intensive.

Geetha et al. [63] developed a system that can recognize and detect faces through facial features. They used a method called Eigenface to extract data from facial images and then trained an ML algorithm called SVM to classify and detect faces. The system can recognize faces from both still images and video frames. The researchers used a dataset of 50 real-time images of students from their institution to test the system, and it achieved an accuracy of about 61%. However, they mentioned that the accuracy could be improved by increasing the dataset size and diversity of the facial images and portraying the idea in the form of graphs.

Slusky et al. [64] did an extensive review and analysis of twenty online proctoring systems to evaluate their security features. This includes assessing multi-factor authentication, blockchain technology, biometrics, and other cybersecurity practices. The study distinguishes between fully automated AI-enabled systems and hybrid systems, integrating automated technology with live proctor oversight. The findings illustrate varied arenas of cybersecurity practices among different online proctoring systems (OPS). Notably, certain systems incorporated advanced security measures like blockchain and biometrics to potentially enhance authentication and authorization processes. The study underscored the effectiveness of these systems in mitigating prevalent security threats and upholding the integrity of online assessments. The research data was thoroughly gathered from a comprehensive review of publicly available information and specifications related to various online proctoring systems, encompassing technical documentation, security protocols, and compliance measures.

Guney et al. [65] proposed a VProctor system designed to create a secure exam environment by blending the hardware and software. It includes facial recognition, gaze and posture tracking, object recognition, and sound classification using various machine-learning libraries and frameworks. It also leverages mobile devices to secure the test environment by requiring 360-degree photos and ID verification, thereby ensuring that the exam setting is free from unauthorized materials and activities. The system effectively detects a range of cheating behaviors, from unauthorized aid to focusing on off-screen locations, using a combination of real-time monitoring and subsequent analysis to ensure compliance and integrity. The results of the study indicate that vProctor reduces the likelihood of cheating in remote exams, aligning online assessment security closely with traditional in-person monitoring standards. The AI capabilities of vProctor are trained and supported by standard datasets like Labeled Faces in the Wild (LFW) for facial recognition and COCO for object detection, ensuring strengthened and generalized performance across varied exam conditions. The system utilizes webcams and microphones as hardware components.

A recent study by Nabil et al. [66] proposes a novel method for detecting cheating in online exams. Instead of relying on video analysis, the researchers developed an Intelligent Examination Monitoring Tool (IEMT) to monitor students' interactions during exams. The IEMT records mouse clicks, keyboard inputs, and other interactions, which are then analyzed using ML techniques such as DT, RF, k-NN, and Naive Bayes. The RF model proved to be the most effective, achieving an accuracy of approximately 91%. This suggests that using behavioral biometrics to identify cheating during online exams is a reliable approach. The study focused on data from students who had taken an online microprocessor exam, which was categorized and analyzed to develop the predictive model.

Kasinathan et al. [67] have highlighted the difficulty of maintaining academic honesty during online exams, citing the lack of proper invigilation as a key issue. To tackle this issue, they have proposed an automated system called "ProctorEx" that monitors students' browser activity and uses camera-based surveillance to identify any suspicious behaviors. The system was developed using the Extreme Programming (XP) software development methodology and merged various modules for automated proctoring, such as face detection, gaze tracking, head pose estimation, and browser activity monitoring. These modules employ ML algorithms like DNN, Dlib, openCV, and image processing techniques to swiftly detect and flag potential cheating behaviors in real time.

Garg et al. [68] address the challenge of upholding academic honesty in online assessments by proposing a unique approach that employs ML to identify cheating behavior during online exams. The approach made use of data collected from two lab-based online assessments using the modified iQuiz tool, integrated with Moodle, which captured detailed interaction data such as responses, response times, scores, response revisions, and tab switch counts. It used feature engineering to extract thirteen features from the assessment log files, based on both student attributes and question-related data. The paper further employs two feature selection algorithms (ANOVA and Mutual Information) and five ML classifiers (LR, SVM, Naïve Bayes, k-NN, and RF) to evaluate the predictive models. The best results were achieved by the RF classifier, using top features selected by the Mutual Information method, with an accuracy of approximately 85%.

Mohammad et al. [69] introduced a system that uses special computer programs called CNN algorithms and image processing software to observe the facial expressions, eye movements, and other behaviors of students during the tests. To make sure the system works well, the researchers used a collection of datasets called CK and CK++, which they split into two parts: 80% to train the system and 20% to test it. Before its usage, the data was cleaned and made extensive with additional samples to enhance its quality.

While other scholars focus on developing real-time online cheating detection systems, Ferdosi et al. [70] investigated cheating behavior in paper-and-pen-based online examinations. They gathered a video dataset, created and recorded within the authors' institutions, of a mock examination that included two types of videos: one showing cheating behavior and the other showing non-cheating behavior. The researchers proposed a model that tracks eyes, lips, and head movements with facial landmarks identified using a media pipe. Movement classification was done using a k-NN classifier, and a 40% threshold was applied to detect cheating behavior and maintain the accuracy of the results.

Varun et al. [21] further contribute to the advancement of techniques aimed at promoting the credibility of online examinations. Their innovative methodology involves collecting data through video recordings and keystroke logs, which are then preprocessed and analyzed using computer vision techniques to extract essential features. By using advanced ML models, the system can successfully detect any cheating behavior. The researchers integrated state-of-the-art facial recognition technology and algorithms such as DeepFace and YOLOv3 for real-time monitoring and authentication.

Desai et al. [71] proposed innovative strategies to enrich online learning experiences. They suggested creating personalized learning paths to counter the lack of personalization in online courses, which commonly led to low completion rates and insufficient

engagement from learners. Their proposed system employs hyper-personalization to customize educational content based on individual learner profiles and utilizes DL models to evaluate attentiveness during learning sessions. The methodology integrates CNNs and RNNs to analyze facial expressions and other indicators of attentiveness. This dual approach not only adjusts the learning material to the learner's requirements but also offers real-time feedback to educators about learner engagement, which can be used to refine and enhance the educational content further.

Simrani et al. [72] proposed an innovative system that used CNN along with K-NN for classification and Principal Component Analysis (PCA) for data analysis. The CNNCDS (Convolutional Neural Networks for Cheating Detection System) system ensures exam integrity through IP address monitoring, session recording, and browser controls, eliminating the need for webcam surveillance. The system achieved an accuracy of 98.5%, surpassing traditional classifiers such as LR, SVM, RF, and Gaussian Naive Bayes. This high performance underscores the efficacy of the proposed system in a real-world educational setting.

Breskina et al. [73] have developed a new prototype that combines advanced features such as emotion and action analysis. The proposed system aims to enhance the functionality of proctoring systems by including analysis of human actions to assess physical activity and monitor hygiene practices when using computers in the learning process, ensuring compliance with personal data protection regulations. The new system is designed to be less intrusive and more supportive of the student's physical and emotional health by providing feedback on posture, physical activity, and stress levels, potentially leading to a more holistic educational environment , .

3.2.1 Advances in machine learning for online exam cheating detection

E-learning platforms have substantially contributed to the development of advanced online examination systems. One such actively adopted method is ML, which offers to enhance the security of these exams by providing sophisticated solutions to prevent cheating. APS, using ML algorithms, analyze behavioral patterns such as facial expressions, eye movements, keystroke dynamics, and browser activities to detect any signs of academic dishonesty. This analysis examines various studies that have utilized ML techniques to strengthen the credibility of online assessments, identifying the most effective approaches based on their performance and application contexts.

3.2.2 Comparative analysis of machine learning approaches for automated proctoring systems

The majority of research on ML for proctoring online exams focuses on textual data analysis and uses techniques such as LSA and SVM to detect plagiarism. It has been proven to be highly accurate, with a success rate of 97%, with the major advantage being its scalability and ease of implementation. Such features make it a popular choice for academic institutions today. Furthermore, methodological innovation is key to the evolution of ML-based proctoring systems. For example, one approach to detecting plagiarism in programming assignments is through ultra-fine-grained repositories, which marks a significant shift from traditional similarity-based approaches. Researchers are also currently exploring the potential of using visual and audio data to detect cheating during exams. For instance, image processing can be employed to monitor students' physical

presence and facial expressions, providing a more dynamic perspective on proctoring. However, the efficacy of these approaches is limited by computational requirements and the complex nature of precisely interpreting visual and auditory signals. Although textual analysis remains a stronghold in the domain, the exploration of visual and audio analyses presents a future where proctoring could transcend conventional boundaries. These novel methodologies, although still in their early stages and facing scalability challenges, offer a more comprehensive dataset for analysis, paving the way for a more integrated approach to upholding the sanctity of online examinations. As online education continues to reach new heights, the fusion of diverse ML techniques holds the promise of developing a stronger, more dynamic, and foolproof system for detecting cheating. The potential convergence of textual, visual, and audio analysis could redefine the standards of academic integrity in the digital age, offering educators and institutions a multifaceted toolkit to safeguard the credibility of online assessments (Tables 7, 8).

3.2.3 Machine learning-based cheating detection: techniques, advantages, and limitations

The implementation of ML in automated cheating detection represents a major advancement towards more intelligent systems in educational technology. Techniques such as SVM, RF, and DT are at the forefront of this transformation, providing powerful analytical capabilities to identify complex behavioral patterns that may indicate cheating during academic evaluations. This section will discuss how ML methods can detect cheating in student behavior and assessments.

Support vector machines (SVM) have gained popularity for automated cheating detection due to their effectiveness in handling high-dimensional data. When combined with techniques like Bag of Words (BOW), Latent Semantic Analysis (LSA), and Stylometry, SVM has proven successful in detecting plagiarism and identifying unethical behavior in online learning environments. It excels in classification tasks and can recognize the optimal hyperplane that separates different classes, even in complex scenarios like cheating detection. Its ability to handle high-dimensional spaces is particularly useful for analyzing intricate patterns in data, such as textual content or user behavior metrics. However, SVM holds certain limitations, including computational intensity, especially for larger datasets, and sensitivity to parameter selection and kernel choice, which can significantly impact model performance. Additionally, interpreting SVM models can be challenging, limiting the ability to extract actionable insights.

The Random Forest (RF) algorithm is a successful tool in ML for detecting instances of cheating. It has proven effective in various studies, such as Ruipérez et al. [51], which used it to identify cheating in MOOCs, and the more recent work by Nabil et al. [66], which employed RF to monitor online exams. One of RF's key advantages is its robustness and accuracy, which allow it to handle large datasets and provide valuable insights into feature importance. Its ensemble approach, combining predictions from multiple DTs, enhances its generalization ability and helps mitigate overfitting. These characteristics make RF an excellent choice for detecting subtle patterns of cheating behavior and identifying suspicious anomalies. On the other hand, RF models can be computationally expensive and complex, making them less transparent and harder to interpret. Additionally, their performance relies heavily on the correct tuning of hyperparameters, such as the number of trees and tree depth.

Table 7 Comparative analysis of behavioral monitoring and cheating detection systems in machine learning-based proctoring

Author	Year	Characteristics	Limitation	Performance
R.S et al. [42]	2015	Heuristic model using webcam and audio cues	High false-positive rate	0.8 FN, 0.13 TN
Li et al. [43]	2015	MOOP framework combining sensor streams and crowdsourcing	Expensive EEG/Gaze sensors	ACD 83.3%, AUROC 90.3%
Asadullah et al. [19]	2016	Detects whispering using spectral and energy features	Limited real-world data	FAR 1%, FRR 3%
Soma et al. [44]	2015	ANN for offline fraud detection via ID card image matching	Limited to printed data fraud	Accuracy 94.12%
Staubitz et al. [48]	2016	Face verification for identity-linked certification	Cannot fully prevent fraud	Not reported
Cote et al. [47]	2016	HMM for summarizing abnormal head pose behavior	Small dataset	Pose accuracy 79.8%
Jalali et al. [20]	2017	Image processing to detect chair vacancy and presence	Relies on static conditions	Accuracy 100%
Ruiperez et al. [51]	2017	RF for detecting CAMEO-style cheating in MOOCs	May overfit; expensive	AUC near 1
Atoum et al. [52]	2017	Multimodal audio-visual detection with webcam and microphone	Dataset-specific results	Accuracy 95.99%
Chuang et al. [50]	2017	LR on head pose and time delay	False alarms possible	Accuracy 75.6%
Soman et al. [53]	2017	HOG+k-NN for gesture-based cheating detection	Limited in crowded scenes	Recall 94.4%, Precision 94%
Northcutt et al. [49]	2017	CAMEO algorithm for multiple-account detection	MOOC-specific data	Not reported
Slusky et al. [64]	2020	Security audit of 20 proctoring systems	Variability in practice	Review-based
Sinha et al. [59]	2020	Full-stack AI with lockdown browser and webcam verification	No real-world eval data	Not reported
Yaqub et al. [61]	2020	Privacy-preserving image hashing-based monitoring	Small-scale testing	Accuracy 75%
Hoque et al. [60]	2020	CCTV + biometric (fingerprint) authentication	Scalability not discussed	Not mentioned
Guney et al. [65]	2021	vProctor with 360° scene capture and ML monitoring	Privacy resistance	AUROC 90.3%
Li et al. [62]	2021	Visual analytics on webcam + mouse data for cheating inference	Context-specific data	Accuracy 88%
Hira Salman et al. [62]	2021	Ensemble DL for COVID-19 diagnosis, Combines VGG16, ResNet50, InceptionV3, Chest X-ray image-based	Limited sample size, Imbalanced datasets, No clinical validation or external testing	Accuracy 88%
Geetha et al. [63]	2021	Eigenface + SVM for real-time face detection	Limited dataset	Accuracy 61%
Nabil et al. [66]	2022	Behavioral biometrics from mouse/keyboard events	No real-world deployment	Accuracy 95.16%
Sanna et al. [23]	2022	Focus on definitions, datasets, techniques, and trade-offs, Discusses group vs. individual fairness	Common ML and DL Approaches	Various
Hussein et al. [74]	2022	Focus on cheating detection in paper-based exams using video surveillance, Introduces a new dataset with 5 classes of cheating actions, Dataset includes real actors and contextual cheating scenarios	Limited to one university and environment, Only frame-level classification used	MSER and SURF+HOG both reached 89%, BRISK: 86%, HOG: 87%
Kasinathan et al. [67]	2022	Browser activity + gaze tracking + CNN (ProctorEx)	High resource demand	Not mentioned
Garg et al. [68]	2023	ML-based log feature extraction in iQuiz-Moodle integration	Only quiz-based	RF accuracy 85%

Table 7 (continued)

Author	Year	Characteristics	Limitation	Performance
Ferdosi et al. [70]	2023	Head/eye/lip tracking in paper-based exams using k-NN	Cannot detect hidden movement	Accuracy 87.5% cheating cases
Varun et al. [21]	2024	CNN/YOLO for keystroke + face detection fusion	No limitations discussed	Real-time performance

Table 8 Comparative analysis of textual and plagiarism-based cheating detection systems

Author	Year	Characteristics	Limitation	Performance
AlSallal et al. [46]	2016	BOW, LSA, Stylometry + SVM for author profiling	English novel-only scope	Not reported
Nagoudi et al. [54]	2018	Arabic plagiarism detection via ML	Language specificity	Plagdet 86.7%, Recall 92.1%
AlSallal et al. [55]	2019	Style-based plagiarism detection using MLP + LSA	Focused on intrinsic analysis only	Accuracy 97%
Qiubo et al. [56]	2019	Code plagiarism via RF + GBDT with behavioral features	Code-only focus	Accuracy 95.9%
Sangalli et al. [57]	2020	Cheating detection via co-occurrence metrics + SVM	Course-specific methodology	Accuracy 95%
Ljubovic et al. [58]	2020	Cloud IDE logging + ML for plagiarism tracking	Specific to IDE use	Accuracy 87.28%
Mohamad et al. [69]	2023	ML + CNN applied to behavior detection (CK, CK++)	No specific limitation noted	Not reported

Decision Trees (DTs) are frequently employed in various applications due to their simplicity and interpretability[8]. DTs provide a clear model structure, making them accessible for comprehending and justifying the decision-making process. Moreover, they are capable of handling various data types with minimal preprocessing requirements. On the contrary, DTs may tend to overfit, particularly with intricate or deep trees, and can be sensitive to small variations in data, leading to disparate outcomes.

R.S. et al. [42] claimed that heuristics-based systems are widely used to monitor behavior using cameras and microphones. Heuristic-based systems are straightforward to implement and rely on predefined rule sets for generating prompt feedback. They are especially helpful in cases where visible, well-defined cheating behaviors are present. However, their inflexibility is a major drawback since they may need assistance in adjusting to novel or unexpected types of cheating. Moreover, there is a significant chance of findings being falsely positive if the heuristics are very strict or lacking.

3.2.4 Advanced feature extraction and analysis for machine learning applications

Feature extraction and analysis are critical elements in ML, transforming raw data into refined variables that significantly improve model performance. The identification of significant features through feature extraction not only boosts accuracy and efficiency, particularly in ML and pattern recognition, but also instills confidence in the data scientists. For instance, in image processing, the extraction of specific features streamlines image classification and object recognition, which provides a clear path for analysis. However, the process of reducing data dimensionality can be quite challenging, particularly with large datasets, and demands careful consideration to avoid losing critical information. Since feature selection has such a critical role in analysis outcomes, it necessitates extensive domain knowledge and trial and error on behalf of ML engineers. In summary, cheating detection systems work by identifying behaviors that may indicate cheating

during online exams. This is done by analyzing different elements, such as how the mouse is moved, how the keys are pressed, and if there are any strange patterns in the answers. Table 9 provides an overview of the common features that are needed in future by AI-powered proctoring systems, their sources, and their intended use. If the system detects any suspicious activity, it will raise an alert flag so that the activity can be investigated. This automated process helps to make the detection process more effective and saves time by reducing the need for constant human supervision. Meanwhile, privacy concerns and the likelihood of false accusations pose potential issues. With cheaters becoming more skilled, continuous system improvements are necessary to remain one step ahead. This complex challenge underscores the need for a delicate balance between technological advancements and ethical considerations.

3.2.5 Devices and tools used in machine learning–based proctoring systems

ML-based proctoring systems rely on a range of hardware devices and software tools to monitor, analyze, and classify student behavior during online examinations. Among the most commonly used devices are webcams, microphones, screen-sharing tools, eye trackers, and keystroke logging software. Each of these tools offers specific advantages but also introduces unique limitations that affect the overall effectiveness and user acceptance of the proctoring system. Webcams are central to visual monitoring and are used to detect head movements, gaze direction, facial expressions, and presence verification. Their primary advantage is affordability and widespread availability across most personal computers and laptops. However, webcams struggle in low-light environments and are vulnerable to spoofing attacks (e.g., using photos or videos). Additionally, they often lack depth perception, which makes it difficult to detect subtle behaviors like signaling or glancing off-screen [38, 40]. Microphones are used to detect background noise, spoken communication, or unusual audio cues. They add an additional layer of behavioral analysis and can flag suspicious sounds (e.g., whispers, third-party voices). The advantage lies in non-intrusive monitoring without requiring visual confirmation. However, microphones are often prone to environmental noise interference, making false alarms common—especially in shared or noisy living spaces [35]. Screen-sharing and keystroke loggers provide real-time access to the student’s digital environment, allowing the detection of application switching, unauthorized browsing, or atypical typing patterns. These tools are instrumental in identifying digital misconduct and function as core elements within ML classification frameworks. The main downside is

Table 9 Key features needed in future for AI-based proctoring systems

Feature	Data source	Purpose
Mouse Movement	Mouse tracking logs	Detecting abnormal navigation patterns and moving too fast
Keystroke Dynamics	Keyboard input	Recognizing unusual typing speed or patterns that suggest cheating
Facial	Webcam / camera feed	Attending to attention, facial expressions, and head pose
Eye-Gaze	Webcam / eye-tracking device	Recognizing deviations in gaze, inattention, or suspicious activity
Screen Activity	Screen capture / system logs	Monitoring tab switching, using applications, or unusual activity
Audio	Microphone input	Recognizing background noise, conversation, or unusual sounds
Answer Patterns	Exam responses	Identifying rapid answering, repeating mistakes, or high similarity to peer answers

the potential invasion of user privacy and resistance from students who perceive such tools as intrusive. Moreover, keystroke dynamics are highly individual-specific and may require a calibration phase, which adds complexity [36]. Eye-tracking devices, such as Tobii or webcam-based gaze estimation systems, are increasingly integrated into ML-based systems to assess attention and detect off-screen glances. These tools provide detailed gaze mapping and behavioral profiling, aiding in real-time suspicion scoring. While their accuracy is impressive in controlled environments, the high cost of specialized eye-tracking hardware and calibration needs limit their scalability. Webcam-based alternatives are less accurate and more susceptible to false detections under movement or lighting variation [32]. Biometric tools, such as fingerprint scanners or face recognition systems, are often employed at the login stage for authentication. While effective in verifying identity and preventing impersonation, their use raises significant data protection concerns, especially under regulations like GDPR and India's Digital Personal Data Protection Act. Moreover, the hardware requirement (e.g., fingerprint readers) is often unavailable on most student devices, reducing practical feasibility [37]. From a software standpoint, tools like OpenCV, TensorFlow, and scikit-learn enable object detection, behavior classification, and model training. These frameworks are open-source, highly customizable, and supported by large developer communities. However, their integration into proctoring platforms requires expertise and extensive testing to avoid bias and ensure real-time responsiveness [39]. For a more detailed breakdown of the devices employed by scholars in developing their proposed systems, refer to Table 10.

3.3 Deep learning–driven automated proctoring system: a CNN-based framework

DL has emerged as a vital technology in online proctoring, particularly for verifying the integrity of remotely administered examinations. These are useful for detecting complex patterns in audio and video data by utilizing neural networks, which allow APS to monitor student activities and detect cheating. Such systems demonstrate high accuracy in detecting behavioral anomalies, including abnormal eye movements, unauthorized

Table 10 Trends in hardware devices across recent research studies

Author	Webcam/camera	Microphone	Mouse	Keystrokes
R.S et al. [42]	✓	✓	×	×
Asadullah et al. [19]	×	✓	×	×
AlSallal et al. [46]	×	×	×	×
Jallai et al. [20]	✓	×	×	×
Ruperez et al. [51]	×	×	×	×
Atoum et al. [52]	✓	✓	×	×
Nagoudi et al. [54]	×	×	×	×
AlSallal et al. [55]	×	×	×	×
Qiubo et al. [56]	×	×	×	×
Sangalli et al. [57]	×	×	×	×
Ljubovic et al. [58]	×	✓	✓	✓
Li et al. [62]	✓	×	✓	×
Geetha et al. [63]	✓	×	×	×
Nabil et al. [66]	×	×	✓	✓
Kasinathan et. al [67]	✓	×	×	×
Garg et al. [68]	×	×	×	×
Mohammad et al. [69]	✓	×	×	×
Ferdosi et al. [70]	✓	×	×	×
Varun et al. [21]	✓	×	×	✓

objects in the exam environment, and the presence of multiple individuals within the camera frame. Many studies [16, 20] have shown how well DL works to identify face features and detect impersonation attempts.

Hendryli et al. [17] classified five groups of suspicious behaviors during tests using the multi-class Markov Chain Latent Dirichlet Allocation (MCMCLDA) technique for prediction and multimodal decomposable models (MODEC) for feature engineering. The identified behaviors include eye contact, document exchanges, hand gestures to interact with other students, observing other students' reactions, and using visual aids. Additionally, one non-cheating behavior is also categorized.

The authors of Cai et al. [75] have identified a significant gap in detecting various cheating behaviors during exam surveillance, which most current methods fail to address adequately. To address this gap, they propose a novel approach using 3D CNNs that can efficiently capture spatial and temporal features from video data. The system preprocesses video into segments, which are then classified independently to detect abnormal behavior. The proposed methodology involves extracting optical flow using Farneback's algorithm and converting this flow into "flow images" that serve as input for the 3D CNN. The model consists of two convolutional layers, pooling layers, and a softmax output for binary classification (normal vs. abnormal behavior). The proposed model achieves an accuracy of up to 89.8% using one method of "flow image" conversion and 87.6% with another. Also, the precision of detecting abnormal behavior reaches up to 86.%, and the recall rate is as high as 95.3% in detecting specific abnormal events that indicate strong performance. A custom dataset is used to collect surveillance video from examination rooms split into predefined regions and includes both positive samples (with abnormal behavior) and negative samples (without abnormal behavior).

Hu et al. [76] have developed an advanced monitoring system that uses a standard webcam to detect potential cheating behaviors during examinations. The system analyzes head posture and mouth state using CNN and ML algorithms. It captures video input, processes it through face detection, and uses performance-based and threshold-based methods to identify abnormal behaviors. The system outperformed conventional proctoring approaches and achieved high accuracy in identifying specific behavioral indicators associated with academic misconduct. It was tested in a controlled online test environment and relies on standard computing hardware and a basic webcam for widespread use in various online examination settings.

S.G et al. [77] proposed a prototype of a biometric M-proctoring system that uses continuous verification techniques to ensure the integrity of the examination process. The proposed model uses a CNN architecture for verification and detection of malpractice during exams, which requires low computation costs and is inexpensive for commercial use. For continuous verification, the user is required to register their frontal, left, and right face pose views in normal and poor lighting.

Nishchal et al. [78] advanced a model to identify cheating behavior in a test through the combination of body language and mood interpretation. The authors employed an open posture tool to analyze posture and an Alexnet architecture based on CNN to identify cheating in exams. There are four different parameters to be detected, such as bending down, staring straight ahead, rotating around, and stretching the arms backward. The ALEXNET model achieved a high accuracy rate of 96% in classifying different types of cheating activities. Emotion analysis also showed an accuracy of 65% in

detecting cheating-related emotions. The combination of all models resulted in a net possible accuracy of approximately 77.8%.

Borse et al. [79] address the urgent need for advanced surveillance methods in examination settings to promote fairness and integrity. Their proposed solution uses 3D CNNs and optical flow techniques to analyze video data for suspicious activities during exams. The approach enables the identification of both spatial and temporal patterns that may indicate cheating, including the detection of hand movements or the presence of additional individuals. The system architecture also incorporates face and hand detection modules to pinpoint interactions between different objects and individuals within the exam room, enhancing the detection accuracy of specific cheating behaviors. To test their methods, the researchers utilized a carefully designed dataset that simulated a variety of cheating scenarios within an examination setting. It consisted of video sequences that captured a diverse range of abnormal behaviors.

Aini et al. [80] discussed abnormal movements during tests using pixel shift and background removal algorithms. They used pixel shift and background removal algorithms on a dataset with various classifications to predict cheating in exams. The background subtraction algorithm was utilized to detect moving objects from a static background, and the Gaussian filter was employed to remove the normal distribution of noise. Additionally, erosion and dilation methods were used to eliminate the foreground area of the images and to enhance the pixels of the images, and contouring was applied to locate and trace the outline of the images. The system utilized a camera connected to a laptop to process images captured during the exam, dividing them into Regions of Interest (ROI) corresponding to each participant's seat. Pixel changes in these ROIs were analyzed, and any suspicious movements were identified on the offset of pixel changes exceeding a certain threshold. This study achieved a testing success rate of 96% for detecting cheating behaviors.

Garg et al. [81] suggested a model to detect students' faces during exams and apply different constraints to prevent malpractice during exams. The authors state that although a typical CNN model provides a better detection rate, the learning process of detection is quite slow. To overcome this issue, they used Haar Cascade along with CNN to improve the performance of the proposed model. Hayatom et al. [18] propose a multi-modal biometric system to verify learner identities and detect cheating behaviors. The model integrates face recognition and keystroke dynamics. Face recognition involves the capturing and processing of face images, while keystroke dynamics analyze keyboard interaction during exams. The dataset is collected from various controlled and uncontrolled environments. The proposed method employs a CNN-based architecture, Visual Geometry Group (VGG) model, for face recognition. VGG pre-trained models are applied to input face images with a resolution of 224x224 pixels to extract feature vectors, which serve as the unique signature of the learner's face. These feature vectors are then used for identity verification within the proposed biometric system.

Saba et al. [16] addressed the challenge of monitoring students during exams by developing an automated system using a DL approach. The authors introduced a novel deep CNN architecture, L2-GraftNet, which integrates features of AlexNet and SqueezeNet. This model was pre-trained on the CIFAR-100 dataset for effective feature extraction. The study also utilized the Atom Search Optimization (ASO) algorithm for feature optimization. Their proposed model demonstrated high accuracy in categorizing student

activities, achieving an impressive 93.88% accuracy with the Fine k-NN classifier, highlighting its potential effectiveness in automated exam proctoring.

Labayen et al. [82] proposed a DL model by authenticating the students using different biometric technologies and a CNN-based proctoring system. Their AI proctoring system is scalable and affordable. They work on different modules, like pose and expression correction in M3L, feature selection, and normalization modules. Light normalizing is done by using the Contrast Limited Adaptive Histogram Equalization (CLAHE) technique. The second part is to recognize the background, which is done by using LSTM, followed by classification performed by extracting all the features using a hybrid CNN+RNN model. The system also monitors the active windows, open and running processes, peripheral devices, browsing history, and copy-paste commands of the respective user.

Jadi et al. [83] advanced a CNN-based model for the online environment during the COVID-19 pandemic. The model uses facial expression and movement detection to analyze and prevent cheating behaviors among students. It achieved an impressive accuracy of 97%. The researchers collected a custom dataset from a 30-minute e-exam, extracting 3600 images from recorded videos for analysis. A similar study on face recognition was conducted by Ganidisastra et al. [84]. They stated that the previous work on face recognition had the limitation of an inefficient AI model against invariant face postures and was unable to detect faces due to poor lighting. To bridge this gap, they proposed a model that incorporates image equalization and SURF features. In their evaluation of a FaceNet model for face recognition, they tested four distinct face detectors: YOLO-face, LBP, MTCNN, and Haar-cascade. The results showed that a deep-learning face detector outperformed the others. However, incremental FaceNet model training led to a 1% smaller dataset size with a faster training time (7% on the Yolo-face and 64% on MTCNN) compared to batch training. The suggested strategy achieved an accuracy rate of 98%, comparable to that of batch training.

Soltane et al. [25] used two devices (a webcam and a microphone) to monitor the activities of examinees during online exams. They employed ResNet-34 for facial, audio, and iris detection. This study was inspired by the work of He et al. [85], but with modifications to the framework, including using fewer layers and half the number of filters. The system also monitors active windows to maintain their integrity. The LFW dataset was used for the study. Kaddoura et al. [23] worked on recorded videos and speeches to proctor the exams. They used three modules to detect cheating: the front camera, the back camera, and speech-related analysis. The study used a CNN and a Gaussian-based DFT model. The authors also used the Online Exam Proctoring (OEP) database, as Atoum et al. [52] did in their study. The proposed model achieved 99.8% accuracy.

Malhotra et al. [24] built an AI proctoring system by maintaining the cost and reliability of the system. They worked on the detection of the emotions of students during exams, the presence of cell phones, documents, books, and other forms of hard materials, along with the presence of other individuals during exams. The system also detects head posture to maintain the credibility of proctoring. To detect emotions, the authors employed a four-layer CNN model with three max pool layers with a Rectified Linear Unit (ReLU) activation layer, three dropout layers with one softmax, and one fully connected layer. They used the FER2023 dataset for this purpose. For head posture detection, they proposed a model using Caffe with the DNN OpenCV module, which first performs face recognition. To detect cell phones, books, and documents, and the

presence of other people, a pre-trained YOLOv5 model with the COCO dataset was used [86].

Li et al. [87] developed a model that uses a front and follower 3D rotated camera to monitor the examinee's activities. The proposed model detects the eye gaze of the examinee in a real-time environment using a 3D gaze motion formula leveraging the TensorFlow CNN model. The approach of the model uses face images that consist of 3D locations of different facial points like the nose, chin, lips, eye corners, mouth corners, etc. The second module of the proposed system incorporates a custom-trained object detection model to detect cheating behaviors like books, sticky notes, cell phones, and calculators. They used a pre-trained SSD MobileNetV2 TensorFlow model and achieved 95% accuracy.

Nafees et al. [31] proposed a CNN model to enhance facial recognition accuracy by employing the CK+ dataset, which has proven more effective than other datasets in facial recognition tasks. The results indicate that CNN, when trained with the CK+ dataset, offers high accuracy in facial recognition tasks, which is crucial for identifying malpractices in online exams. The study also highlights the potential for future research to develop real-time APS using CNN. Hardware devices such as webcams are crucial in capturing facial data, which the CNN model then processes to detect potential fraudulent activities during examinations.

Liu et al. [88] proposed an offline AI-based proctoring system to check and monitor the cheating behavior of students in an offline environment. The model utilizes a 2-longitudinal-stream CNN-based architecture. A comparison between human proctoring and the proposed model was conducted to evaluate the accuracy of the model's performance. The result indicated that the model effectively detected instances of cheating with minimal performance requirements. The model was trained and tested using a custom dataset consisting of six videos of school examination environments. In contrast, Tejaswi et al. [89] built an AI proctoring system to detect cheating behavior in the online environment. The system focuses on analyzing three specific behaviors: students' mouth opening, eye movement tracking, and examinee recognition. The model is based on CNN architecture, utilizing the Inception ResNet base CNN model, GoogleNet, mouth aspect ratio techniques, and hourglass modules to enhance accuracy. The system was trained on a dataset of 200 videos, with 5% focused on instances of malpractice, 20% on suspicious student behavior, 25% on examinee's behavior, and 25% on examinees seeking help from others.

Potluri et al. [90] also worked in a real-time environment for proctoring. The author worked in a live examination environment to analyze and detect the cheating behavior of examinees. They proposed a CNN-based model for multiple-person detection, spoofed facing, and estimation of examinees' head postures. The proposed method achieved an accuracy of 87%.

Yulita et al. [91] built an Indonesian language-based automated exam proctoring system. They used video recordings of exams from Michigan State University and preprocessed them by extracting frames. The model was trained using a 2D-CNN framework. The first layer of CNN involved 2D convolutional processing with a 1x1 kernel size, followed by a 2D max pooling layer with a 2x2 size. Subsequently, the data was flattened into 400 dimensions to connect all the data from the previous layers to the activated layers. The data is then passed to a thick layer that receives the feature mapping.

Zaffar et al. [92] built a hybrid model combining CNN and RNN frameworks to detect four elements: tracking and identifying examinees, identifying suspicious activities, marking attendance, and checking cheating behaviors. YOLO-v7 and deep sort were used for student tracking, the Inception v3 model was used for feature extraction of suspicious activities, and STMP was used to generate alert notification emails.

In their proctoring system, Kulshethra et al. [93] use face detection, gaze detection, mobile phone detection, noise disturbance, and headphone detection. The authors employed the MTCNN model to monitor the students' cheating behaviors in both online and recorded environments.

Saravanan et al. [94] proposed an automated method utilizing computer vision to address cheating during physical exams, a persistent issue in educational settings. Conventional means of invigilation may sometimes lag due to constraints in monitoring and ensuring the integrity of the examination process. They proposed a system that uses CCTV cameras along with the YOLOv3 algorithm for precise object detection and the ShuffleNet architecture for computational efficiency to detect suspicious actions during examinations. The system includes camera networks for capturing images, behavior classification, real-time monitoring, and an alert system that guarantees the identification of cheating behavior while adhering to ethical standards. In their study, the system reached an 88.03% rate in detecting cheating in a classroom environment. The experiment utilizes a dataset that captures head and neck movements to identify cheating behaviors and demonstrate the more accurate performance of the developed model compared to existing techniques. The authors highlighted future areas of improvement such as accuracy, adaptation to new cheating methods, multi-modal approaches, privacy-preserving techniques, real-world deployment and validation, interoperability, bias mitigation, scalability, ethical and legal frameworks, and education and awareness.

Mohajar et al. [95] address the challenge of monitoring cheating during e-exams by summarizing exam videos to focus only on scenes that potentially include cheating. Existing methods struggle with large datasets and accuracy issues when new background scenes are introduced. The researchers identified the need for an efficient video summarization technique that can handle new scenes without extensive pre-training and introduced a deep-learning method to tackle the issue. The methodology includes the detection of examinees and body parts using YOLOv5 and OpenPose libraries, respectively, followed by a selective summarization process based on structural similarities and the predefined significance of different body parts. The method's objective is to help decrease the computational load while maximizing precision and speed in comparison to other methods. It can achieve up to a 70% reduction in video length while maintaining a high level of accuracy. The proposed method was initially trained using COCO and Penn-Action and then tested on a custom dataset containing 60 h of video from 120 participants. This ensured efficiency in identifying relevant behaviors and summarizing them appropriately. The experiment used standard video recording equipment like a webcam, and the videos recorded were processed on a PC with an Intel Core i7-7500U and 8GB of RAM.

Verma et al. [96] developed a multi-modal, AI-based system based on DL methods to improve the robustness of online exam proctoring. The system incorporates emotion detection; head pose estimation, malicious object detection, face spoofing detection, and mouth opening detection. These modules utilize CNN and other complicated algorithms

to analyze the videos captured during the examination. The study used real-time video data from standard online tests or exams to highlight cheating behaviors identified by the different AI-based modules in the system. The system has been tested for various functions, demonstrating promising capability in effectively detecting cheating behaviors. A webcam is used as the primary hardware in the system. A detailed comparative analysis of the literature review of CNN models is shown in Table 11.

3.4 Devices and tools used in a CNN-based proctoring systems

Based on a comprehensive analysis of existing research, it has been observed that scholars utilize an array of tools and techniques within the CNN framework to address the issue of detecting cheating in both offline and online exams. These methods comprise the use of cameras, such as CCTV cameras during real-time exams and laptop cameras during online exams. To establish these proctoring detection systems, various algorithms are employed to monitor online exams using the cameras situated on the front and back of the laptop. A detailed breakdown of the devices employed by scholars to develop their proposed systems can be found in Table 12.

3.5 Automated proctoring systems based on diverse CNN architectures

This section highlights the CNN methods used to detect the cheating behaviors of the students. It has been observed that the CNN framework enhances APS, but deep CNN architectures can be computationally complex and prone to over-fitting. While they offer high accuracy and robust feature extraction, they may require substantial computational resources and abundant labeled data for training, posing challenges in resource-constrained environments. This section discusses the working process with pros and cons of the employed architecture and outlines the advantages and disadvantages of each model.

3.5.1 AlexNet

AlexNet is known for its high accuracy in image classification due to its deep architecture that allows learning from large datasets and extracting features efficiently. Its adaptability to various applications and transferability to novel situations make it a significant asset. However, its complexity can make it difficult to interpret, and it may be prone to over-fitting if not appropriately regulated. Additionally, its dependence on extensive datasets and the need for expertise in tuning make it challenging to use in projects with limited resources [97].

3.5.2 GoogleNet

The novel architecture of GoogleNet, particularly its Inception versions, brought significant improvements in the accuracy and effectiveness of neural networks. It uses inception modules to perform convolutional operations simultaneously at various scales, which makes it efficient in utilizing computational resources while maintaining high accuracy. This model reduces the number of parameters, which makes it less likely to be overfit and more versatile than previous models like AlexNet. Its inception layers capture detailed features at different levels, which makes it highly effective in image classification tasks. However, the complexity of this architecture can pose challenges for those scholars who try to modify or fully understand the model's inner workings. It takes a lot of processing power to train GoogleNet from scratch, and fine-tuning its hyperparameters

Table 11 Comparative analysis of automated proctoring systems using a CNN framework

Author	Year	Characteristics	Limitations	Proposed model	Performance metrics	Datasets
Hendryli et al. [17]	2016	Advanced detection of various abnormal behaviors using 3D CNNs; robust model with high accuracy and recall	Specific hardware details not disclosed; may require high computational resources	3DCNN	Accuracy up to 89.8%, Precision 86.5%, Recall 95.3%	Custom-made dataset of examination surveillance videos
Cai et al. [75]	2016	Efficiently detected cheating during examinations	Difficulty in differentiating similar-colored clothing	CNN architecture: MCMCLDA (Multi-class Markov Chain LDA)	Accuracy: 80%	Simulation videos from CCTV cameras
Hu et al. [76]	2018	Automation of cheating detection with high accuracy; less reliance on manual monitoring	Requires initial calibration and setup for accurate results	CNN-based head pose and mouth state analysis	Not mentioned	Custom dataset of 30 simulated test videos
S.G et al. [77]	2019	Continuous verification registration method to ensure the integrity; proposed prototype is designed for mobile based exams	The proposed model is in the prototype stage and hasn't been implemented	CNN architecture	Not mentioned	Captured live video recording
Nishchal et al. [78]	2020	An efficient solution for cheating detection	Dependence on dataset quality; Only Applicable to offline exams; limited to lateral camera positioning	CNN architecture: Open pose, ALEXNet	Accuracy: 77.8%	Emotion analysis: FER-2013, CK+; Face recognition: custom dataset; Posture detection: Open pose, Radboud Faces Database
Aini et al. [80]	2020	Real-time monitoring of participants during exams	Limited to detecting movements within ROIs	ML, data analysis model	Accuracy: 96%	Custom: Camera-captured images
Garg et al. [81]	2020	Used to proctor randomized MCQ tests	Only face recognition is the focus of this study	Haar cascade CNN	Accuracy: 93%	Not mentioned
Hayatom et al. [18]	2020	Monitoring of keystroke dynamics during exams	There's a need for a larger dataset and the impact of varying environmental conditions on performance	VGG model	ERR rate	Collected images from controlled and uncontrolled exam environments
Borse et al. [79]	2020	Advanced detection of dynamic cheating behaviors, reduces manual monitoring errors	High computational resources may be required	3D CNN with optical flow	Not mentioned	Custom dataset simulating examination cheating
Saba et al. [16]	2021	High accuracy in student activity categorization using deep CNN	N/A	CNN architecture; L2-GraftNet	Accuracy: 93.88%	Pre-trained with CIFAR-100, CUI-EXAM(2268 images)

Table 11 (continued)

Author	Year	Characteristics	Limitations	Proposed model	Performance metrics	Datasets
Labbeyan et al. [82]	2021	Fully automated commercially available and affordable E-proctoring system	Undesirable deviations due to variations of face pose and light	Authentication of biometrics with CNN+RNN	Precision- 99%, Recall-74%	Live/real-time proctoring
Jadi et al. [83]	2021	Innovative CNN-based system for effective cheating detection in online exams	Limited dataset	CNN	Accuracy: 97%	The custom dataset used 10 students' webcam images of 30 min E-exam
Ganidisasthra et al. [84]	2021	Overcome the invariance of face recognition due to poor lighting and bad face pose	N/A	CNN-FR, LB, MTCNN	Accuracy: 98%	Videos of lecturing sessions
Soltane et al. [25]	2021	N/A	Lack of detailed validation results to demonstrate the effectiveness of the proposed system	CNN: ResNet-34	Accuracy: 99.38%	LFW dataset
Kadoura et al. [23]	2022	Achieved high accuracy	N/A	CNN and Gaussian-based DFT	Accuracy: 99.8%	OEP database
Malhotra et al. [24]	2022	Works on cell phones and documented materials as well	Human intervention may be used for the detection of other types of cheating behaviors	CNN, DNN, YOLOv3	Accuracy: 90%, Accuracy: 75%	FER2013, COCO
Li et al. [87]	2022	3D rotation of eye-gaze tracking	Lack of discussion on the potential impact of environmental factors like poor lighting, background clutter, etc	3D gaze detector using CNN, SSD Mobilenet	Accuracy: 95%	Real-time environments
Smirani et al. [72]	2022	Innovative approach to online exam proctoring without webcam use, high accuracy in cheating detection	Dependency on specific software compatibility and student compliance with system requirements	CNNCDS using CNN for behavior analysis, K-NN for classification, PCA for data analysis	High accuracy with CNNCDS at 98.5%, superior to traditional methods	Exam session data, IP logs, and browser activity from real exam environments
Nafrees et al. [31]	2022	Provide a comprehensive review of ML techniques for facial recognition in online proctoring; highlighting the superiority of CNN with the CK+ dataset	The study was limited by the availability of recent research and funding to access certain journals; potentially overlooking some current developments	Use of CNN for facial recognition to detect examination fraud	Not mentioned	CK+ dataset used for training the facial recognition model

Table 11 (continued)

Author	Year	Characteristics	Limitations	Proposed model	Performance metrics	Datasets
Mohajer et al. [95]	2023	High processing speed; efficient summarization	Not mentioned	DL-based video summarization using YOLOv5 and OpenPose, modified SSIM	Accuracy: Up to 70% video summarization achieved	COCO, Penn-Action, custom exam footage
Saravanan et al. [94]	2023	High accuracy, efficiency, real-time alerts, adaptability, ethical compliance	Privacy concerns, false positives, evolving cheating techniques	YOLOv3 with ShuffleNet architecture	Accuracy: 88.03%	Dataset capturing head and neck movements for cheating detection
Liu et al. [88]	2023	Real-time environment videos are used	Limited content and size of the dataset	CNN-based 2 longitudinal stream	Accuracy: 89%	6 videos of school-based scenarios
Tejaswi et al. [89]	2023	Provide higher results on testing datasets	Failed to adapt to open book exam environment; Failed to adopt students who are autistics	DesNet, GoogleNet, Resnet	Accuracies: 91.4%, 91.2%, 90.7%	200 videos Custom dataset (YouTube, Proctor dataset, Wild dataset)
Potluri et al. [90]	2023	Detection of malpractice in a real-time environment	The project is restricted to Video Processing	CNN	Accuracy: 87%	Live E-exam
Yulita et al. [91]	2023	Indonesian language-based web app	Rely on video frames that may not detect another form of cheating behavior	2D CNN	Accuracy: 84.4%	Online exam videos from Michigan University
Kulshes-thra et al. [93]	2023	Detects mobile phones, earphones and other objects	The paper did not provide performance metrics	CNN, MTCNN	Not mentioned	100 recorded images as well as online environments
Zaffar et al. [92]	2023	Real-time email alerts	Audio analysis and eye-tracking has not been covered	CNN+RNN, inception V3 model	90.67%	Real-time videos
Verma et al. [96]	2024	Integration of multi-modal detection features; Automated real-time analysis; AI-driven approach for higher accuracy	Specific performance metrics not detailed; may require high computational resources	CNN model	Not mentioned	Video data from online examinations

can help it operate at its best. Despite these difficulties, it is a useful tool in DL techniques because of its capacity to strike a compromise between high performance and computational balance [97].

3.5.3 Inception V3

Inception V3 is an advanced version of Inception models with improved performance and efficiency. It uses factorization techniques to reduce computational costs while maintaining depth and width. It is highly versatile in handling various data scales and has state-of-the-art feature extraction capabilities, which makes it accurate for complex tasks. Inception V3 is adaptable to different computing environments and can function

Table 12 Trends in hardware devices across recent research studies

Author	Front camera	Back camera	CCTV	Keystrokes	Audio/speech
Saba et al. [16]	✓	✓	×	×	×
Hendryli et al. [17]	✓	✓	×	×	×
S.G et al. [77]	✓	×	×	×	✓
Nishchal et al. [78]	✓	×	×	×	×
Aini et al. [80]	×	×	×	×	×
Garg et al. [81]	✓	×	×	×	×
Hayatom et al. [18]	✓	×	×	✓	×
Labayen et al. [82]	✓	×	×	✓	×
Jadi et al. [83]	✓	×	×	×	×
Ganidisastra et al. [84]	✓	×	×	×	×
Soltane et al. [25]	✓	×	×	×	✓
Kaddoura et al. [23]	✓	✓	×	×	✓
Malhotra et al. [24]	✓	×	×	×	×
Li et al. [87]	✓	✓	×	×	×
Liu et al. [88]	✓	✓	×	×	✓
Tejaswi et al. [89]	✓	×	×	×	✓
Potluri et al. [90]	✓	×	✓	×	×
Yulita et al. [91]	✓	×	×	×	✓
Zaffar et al. [92]	✓	×	×	×	✓
Kulshestra et al. [93]	✓	×	×	×	✓

constructively on powerful servers and mobile devices. However, adjusting and fine-tuning the model without a firm understanding of its architecture is complicated. Additionally, training Inception V3 from scratch is resource-intensive, and its "black box" nature makes it difficult to comprehend how decisions are being made [97].

3.5.4 ResNet

ResNet is a DL architecture that addresses the problem of vanishing gradients encountered by earlier models. It introduces residual blocks that facilitate the direct propagation of gradients through shortcuts, enabling the training of networks with hundreds of layers. The model improves the network's ability to learn complex patterns without a proportional increase in training difficulty or resource demand, which makes it highly effective in various applications. However, training ResNet from scratch can still be computationally intensive, and its complexity can make it challenging to understand and modify. Despite these challenges, ResNet's ability to expand the possibilities of DL while having lower demands on resources makes it an excellent choice for advanced applications [97].

3.5.5 Other techniques (Markov chain latent Dirichlet allocation, keystroke dynamics)

Various methods are used to integrate different data types, such as text, pictures, and behavioral data. These methods include Markov, Chain Latent Dirichlet Allocation, and keystroke dynamics, which provide detailed information for analysis and detection. Other techniques, like body language and keystroke analysis, are more suitable for deployments in various settings due to their non-intrusiveness, as they do not require additional hardware or sensors and address privacy concerns. Exam data can be processed in real time using approaches such as pixel shift and background removal algorithms to facilitate early detection and necessary actions. Additionally, for identity verification, traditional biometric authentication techniques, such as fingerprinting

and eye scanning, are replaced with privacy-preserving techniques, such as keystroke dynamics and face recognition. Furthermore, Latent Differential Analysis (LDA) and pixel shift algorithm methods are flexible and may be customized to certain situations or settings to meet a range of test conditions and circumstances. Some non-image-based techniques might be less feature-representative than the CNN framework, which could affect the accuracy of detection of cheating behaviors, especially in complex scenarios. On the other hand, certain methods, which highly rely on biometric signals or behavioral data, are not reliable and have less robustness because of data quality issues. Such methods can be impacted by noise and environmental variability. Different solutions, such as a multimodal biometric system, could require extra hardware or specialized equipment for data collection and processing. It could raise the cost of implementation and create logistical difficulties. It is also observed that some non-image-based approaches can be less generalizable for different contexts of cheating behavior and activities. Such approaches are suited only for certain domains or applications, which could restrict their wider applicability.

3.6 Automated cheating detection based on LSTM and RNN

Long Short Term Memory (LSTM) and RNN are essential technologies for automated cheating detection. They are advanced deep-learning models capable of monitoring and analyzing online exam behaviors. These models can process and predict time-series data, providing valuable insights into sequential and temporal patterns of test-taker actions. As a result, they are widely used in various proctoring solutions to enhance security and operational efficiency in response to the significant increase in online education [98, 99]. RNNs capture temporal dependencies in behavioral sequences such as gaze shifts, tab switches, and typing patterns, while LSTMs go further by overcoming vanishing gradient issues, making them more suitable for long-term pattern recognition.

Arinaldi et al. [100] developed a cheating detection system that uses video feeds to identify illicit gestures and behaviors during exams. The automated system combines gesture recognition and natural language generation to document instances of cheating. The methodology involves a gesture recognition model using 3D Convolutional Neural Networks (3DCNN) and XGBoost, as well as an LSTM network for generating textual descriptions of these gestures. The study utilized a dataset of 71 videos of students engaging in various forms of cheating.

The study of Tiong et al. [26] developed a technique to prevent cheating during online assessments. The researchers created an "e-cheating intelligent agent" that includes an IP detector and a behavioral monitor to detect misconduct. The system assigns randomized assessment sets based on IP addresses and uses a DenseLSTM network to analyze patterns in candidate responses and detect any irregularities. The system achieved a classification accuracy of up to 90.7% on a new dataset called PT Behavioral Database, which includes behavioral data from online assessments conducted under various conditions to simulate a real-world environment.

Kadyrov et al. [101] addressed the issue of detecting automated reading during online oral exams where reading is not permitted. The researchers created a dataset of short video clips simulating an online exam environment and used video augmentation techniques to enhance the diversity and robustness of training data. They employed two DL models, the Convolutional Long Short-Term Memory (ConvLSTM) architecture and the

Long-term Recurrent Convolutional Network (LRCN), to analyze spatial and temporal data, effectively interpreting subtle reading movements captured in the video frames. The study found that the LRCN model achieved flawless 100% accuracy even under augmented training conditions. This technology could be used in real-world educational settings to prevent academic dishonesty, using commonly available webcams, making it practical and feasible without requiring specialized equipment.

Tzeng et al. [102] proposed a novel approach that employs AI and eye movement detection technology to analyze online exam behaviors and detect potential cheating. The approach was to tackle a common issue in online proctoring by using eye movement patterns to assess the likelihood of cheating during remote exams. The method involved a specialized AIPS that tracks eye movements via a webcam. By combining LSTM networks, the system analyzed temporal variations in gaze patterns and detected anomalies in test responses. The study involved the creation of a unique dataset by capturing student eye movements during mock exams, which was subsequently used to train and validate the AIPS model. The AIPS demonstrated high accuracy in detecting deviations in eye movement. The study used a customized dataset from a controlled group of participants during online tests.

Felsing et al. [103] addressed the lack of efficient, resource-light methods for proctoring online exams in settings with limited resources. They introduced a methodology that involves a suspicious behavior detection module using pre-trained models for action recognition and a visualization module. The system achieved a 78.9% true detection rate for suspicious actions and reduced the time required by invigilators to monitor exams effectively. The dataset used includes webcam footage from twenty-four test-takers, and the primary hardware used is the laptop webcam to support cost-effectiveness and easy deployment in resource-constrained settings.

Tang et al. [104] conducted a study on the detection of atypical test-taking behaviors using LSTM networks. The research aims to address the limitations of current methods in detecting such behaviors, which are often based on simple statistical indicators or proxies that need more detailed contextual analysis. The study introduced a Model Agreement Index (MAI) as a means to measure the accuracy of the LSTM model's predictions of test-taker behaviors. The researchers analyzed clickstream data, which records all actions taken by test-takers within the testing software. This data was then used with an LSTM network to predict future actions based on historical data. The effectiveness of this approach was assessed using the MAI, which quantifies the typicality of test-taking actions and identifies deviations from predicted patterns. The study found that the LSTM model generally performed well, and certain atypical behaviors were linked to inefficiencies in the testing interface. For example, problems with tool usage led to wasted time for test-takers. This insight not only aids in detecting cheating but also informs improvements to the overall user interface of testing software. The dataset used in the study included comprehensive clickstream logs from a statewide summative assessment for Grade 8 students, providing a strong foundation for analyzing and predicting test-taking behaviors. A comprehensive analysis of the literature review of LSTM and RNN models for cheating detection is shown in Table 13 below.

Table 13 Comparative analysis of automated proctoring systems using LSTM and RNN framework

Author	Year	Characteristics	Limitation	Proposed model	Performance metrics	Datasets
Arinaldi et al. [100]	2017	Automated real-time system; high accuracy in gesture recognition	Limited to pre-defined gestures; might not generalize well	3DCNN and XGBoost for gestures; LSTM for text generation	Gesture recognition 81.11% accuracy; Language generation up to 96.6% word accuracy	71 cheating videos
Tiong et al. [26]	2022	Advanced AI-driven methods for detecting e-cheating; high classification accuracy	Potential limitations in adaptability to different testing formats and environments	Integration of IP detection and DenseLSTM for behavioral analysis	Accuracy: 90.7%	PT Behavioural Database; collected from online assessments
Kadyrov et al. [101]	2022	High accuracy in detecting reading; uses standard webcam hardware	Model specificity to reading detection might limit broader application without further adaptation	ConvLSTM and LRCN models with analyzing spatial-temporal features	Accuracy: 100%	Custom dataset simulating online exam conditions enhanced by video augmentation
Desai et al. [71]	2022	Enhances online learning through tailored educational pathways; utilizes advanced DL techniques to monitor and improve learner engagement	Implementation complexity and the need for robust data privacy measures; Relies on the accuracy of the ML models for attentiveness detection	Hyper-personalized learning paths based on individual assessments; attentiveness is measured via CNN and RNN by analyzing facial expressions and behaviors	Not specified in detail; effectiveness implied through the described system capabilities	Utilized both generated and real-time data for training the DL models
Felsinger et al. [103]	2023	Effective detection of misconduct; Low-cost hardware requirement; Minimal invigilator intervention needed	Limited by resource constraints for real-time analysis	RNN	True detection rate of 78.9%	Online exam dataset; twenty-four test takers who replicate real-world actions
Tzeng et al. [102]	2023	Innovative use of AI and eye-tracking for exam proctoring; high potential accuracy	Limited by the scope of eye movement data and potential variations in individual eye movement behaviors	AI-based system utilizing LSTM for analyzing eye movement patterns in exams	Significant ability to detect anomalies in eye movements; specific accuracy metrics are not detailed	Custom dataset comprising recorded eye movement data during mock exams
Tang et al. [104]	2023	Provides a novel quantitative measure (MAI) for behavior prediction accuracy	Limited discussion on the scalability to other forms of testing beyond K-12 education	LSTM architecture	Not mentioned	Clickstream data from statewide summative tests for grade 8 students

Table 13 (continued)

Author	Year	Characteristics	Limitation	Proposed model	Performance metrics	Datasets
Nurpeiso-va et al. [105]	2023	Developed a cost-effective, university-owned Proctor SU system; Compares results with 3 commercial systems (OES, Oqylyq, Aero); Open-source and integrated with Moodle	Slower performance under poor internet, Lacks 360° camera feature (unlike commercial systems), No fingerprint or iris scanning (due to cost), Still under licensing process	Architecture: B/S model using Python (Django), PostgreSQL, AI models: YOLOv3, CNN, R-CNN, Face and voice recognition, screen monitoring, alerts, exam logs	Accuracy 98.95%	5000 test face images from students, Includes various head orientations and lighting conditions, System trial with 120 students and 10 teachers

3.6.1 Comparative analysis of LSTM and RNN for detecting cheating behaviors

Online assessment authenticity is a crucial issue in digital education. With the use of LSTM and RNN in artificial intelligence, cheating behaviors can now be adequately detected. These technologies analyze complex patterns in student behavior and test-taking methods to identify any anomalies that may indicate dishonest practices. LSTM models can detect irregular examinee patterns by capturing temporal dependencies in behavioral data. Tang et al. [104] utilized LSTM networks with clickstream data to accurately forecast students' behaviors through an MAI. This approach can pinpoint unusual behaviors and uncover inefficiencies in testing interfaces that may encourage cheating. LSTM models offer more in-depth insights compared to traditional statistical indicators due to their ability to learn from sequential data. However, the study had limited scalability and needed to determine its applicability beyond K-12 educational settings. Additionally, Tiong et al. [26] made LSTM even better by adding a DenseLSTM framework. This framework watches IP addresses and looks at user behavior to find cases of e-cheating. Their system boasts an impressive level of accuracy in classification, suggesting its potential for practical implementation. However, the adaptability of the model to diverse testing conditions and settings could potentially hinder its widespread applicability. RNN, including ConvLSTM and LRCN variants, have shown potential in scenarios involving the interpretation of spatial and temporal data. Kadyrov et al. [101] utilized ConvLSTM and LRCN models to detect subtle indicators of reading during online oral exams, achieving 100% accuracy under augmented training conditions. The use of typical webcams for data collection suggests the practical possibility of implementing this system on a larger scale. Nevertheless, the models' focus on identifying reading-related cheating may only allow their applicability in broader academic cheating contexts with additional adjustments. LSTM and RNN models are effective for detecting cheating in online assessments. LSTM is suitable for handling large data sequences, while RNN is ideal for scenarios where spatial and temporal dynamics are crucial. However, LSTMs may require extensive training data and need assistance with scalability, while RNNs might be too specialized. Ultimately, the choice between the two models depends on the specific requirements of the cheating detection task at hand. Table 14 represents the studies which use different hardware devices to develop cheating detection systems.

Table 14 Hardware utilization in LSTM- and RNN-based automated cheating detection studies

Author	Webcam/camera	Microphone/speaker	Mouse/clickstream	Keystrokes
Arinaldi et al. [100]	✓	×	×	×
Tang et al. [104]	×	×	✓	✓
Tiong et al. [26]	×	×	×	×
Kadyrov et al. [101]	✓	×	×	×
Tzeng et al. [102]	✓	×	×	×
Desai et al. [71]	×	×	×	×

3.7 Automated cheating detection based on IoT and emerging technologies

The IoT technology is cost-effective, adaptable, and user-friendly for automated cheating detection in online exams. It employs basic hardware, such as webcams, microphones, and internet connections, to create these systems. The advancements in AI proctoring systems are particularly important during the COVID-19 pandemic, when digital learning is being actively adopted across the globe [106]. Several studies have proposed architectural and functional frameworks combining sensor networks with AI analytics to detect anomalies in student behavior during online exams [107, 108].

Shevale et al. [109] have developed a web-based application aimed at enhancing accessibility for disabled users during online examinations. The application, built on the MERN stack, integrates Python libraries for text-to-speech and speech-to-text functionalities, providing a coherent and secure proctoring solution. The system's utility is highly stressed in the paper, particularly in addressing the need for strengthened security of online exams and proctoring due to the COVID-19 pandemic. One of the system's distinguished features is its voice navigation capability, which makes it easier for users with disabilities to interact with the interface. Students can easily navigate through the examination system using voice commands, significantly improving usability for individuals with physical disabilities. The system only requires minimal hardware, including a standard webcam, microphone, and internet connectivity.

Ho et al. [110] researched remote proctoring and found limitations in the existing methods. They developed a new solution called RAPID that uses a Raspberry Pi Zero W device to monitor test-takers' activities during an exam. The device uses advanced security techniques to detect cheating and prevent tampering. RAPID collects data on the test-taker's computer, such as network traffic and running processes, which are then sent to an invigilator's portal for analysis. The Raspberry Pi Zero W device is used to collect data and interact with the test-taker's computer to ensure the examination process's integrity.

Nguyen et al. [111] focused on improving the integrity of online exams through a Proctoring of Online Test (POT) system. The system relies on IoT technology and AI algorithms to authenticate users and detect cheating behaviors. It uses a dual-camera setup for real-time monitoring and analysis of candidates' actions, along with AI techniques for identifying fraudulent activities. The findings show that the POT system can detect cheating with over 85% accuracy. To train the AI algorithms meticulously, a dataset of video recordings was curated from online exams. These videos were carefully segmented into one-second clips and meticulously labeled as fraudulent or non-fraudulent, creating a comprehensive and reliable training dataset. The hardware used in this study, including the NVIDIA Jetson Nano and Nano IMX219-160 cameras, were key elements in capturing and processing data from the exam environment.

La Roca et al. [112] examine the impact of online proctoring on students' satisfaction and perceptions of a MOOC. The research uses a mixed-method approach to analyze data collected from surveys and exams administered to students enrolled in a digital marketing course on edX. The methodology investigates students' access to technical resources, their emotional states before, during, and after proctored exams, and their perceptions of the online proctoring process. The findings indicate that while most students have the necessary resources, a significant minority do not, which can limit their performance in a proctored environment.

Plochaet et al. [113] investigated the use of video anomaly detection technologies for monitoring academic fairness during online exams. The researchers utilized a combination of video anomaly detection, voice activity detection, and person detection models to identify cheating behaviors. The study involved training the anomaly detection model on 'normal' behaviors and testing it against 'abnormal' behaviors indicative of cheating. The initial results showed a true positive rate of 0.72 and a false positive rate of 0.31 for the video anomaly model. A basic hardware setup consisting of webcams and microphones with a fish-eye lens was used to detect voice and humans, thus improving cheating detection.

Atabay et al. [114] conducted a study on the effectiveness of BeatGAN, which is a generative adversarial network designed for time-series anomaly detection. The network performed well in identifying sequence anomalies in skeleton-based motion data from exam participants, particularly in detecting cheating behaviors. However, it struggled with event anomaly detection. The study utilized a semi-supervised learning approach due to the rarity and variability of cheating behaviors, and the dataset included skeleton features extracted from electronic exam videos. It concluded that advanced ML models do have the potential to enhance monitoring in remote exams, but improvements are necessary to reduce false positives and detect subtle cheating methods.

Bommireddy et al. [115] have identified a noteworthy challenge in effectively proctoring online exams, which is to oversee multiple students simultaneously and uphold the integrity of the testing environment. To address these challenges, they introduced a system that utilized AI technologies to monitor test takers through video and audio inputs. The methodology involved using AI to detect multiple types of potential cheating behavior, including head pose estimation, object detection, and voice recognition. The system processed video streams to identify unauthorized objects like mobile phones or books, analyzed the direction of a student's gaze, and recognized voices to detect if the examinee was communicating with others. Standard webcams and microphones were used to capture video and audio, and AI models were trained on video and audio recordings from actual exam settings. The system proved to effectively identify cheating behavior, reducing the need for human proctors.

The comprehensive analysis of the literature review of IoT devices for cheating detection is illustrated in Table 16 as shown below. Online assessments rely on automated cheating detection systems to ensure that online exams are conducted without any dishonest activities intervening in their integrity. IoT and other technologies have led to advanced tools that monitor and analyze test-taker behaviors in real time. These tools integrate IoT devices, advanced imaging, and ML algorithms to prevent cheating in online environments. They provide real-time monitoring and security during online exams. For instance, Nguyen et al. [111] achieved over 85% accuracy in identifying

cheating by using a dual-camera setup with AI algorithms to authenticate users and detect fraudulent activities. However, the method requires extensive preparation from candidates to ensure accuracy and witness challenges such as varying lighting conditions. Ho et al. [110] used a Raspberry Pi Zero W to monitor test-takers' activities through network traffic and system processes. This approach, called RAPID, shows the potential of IoT in low-cost and efficient proctoring solutions without additional software. However, it is currently limited to Windows OS, so more compatibility is needed to make it more widely applicable. It is a renowned fact that, in current times, AI and ML models are actively being implemented to detect cheating behaviors in exams. Bommireddy et al. [115] have developed a system that monitors students through video and audio using AI techniques such as head pose estimation, object detection, and voice recognition. This technique reduces the need for human proctors and improves the integrity of testing, but its effectiveness is limited to specific cheating behaviors. Atabay et al. [114] have explored the use of BeatGAN, a generative adversarial network, for detecting anomalies in skeletal movement data, but it is limited in its efficacy in detecting event anomalies. When assessing technological solutions, IoT devices often stand out for their ability to collect and monitor real-time data, making them an effective option for continuous observation. However, these systems can be limited by hardware requirements and challenges in certain operational environments. In contrast, AI and ML approaches, like those developed by Bommireddy et al. [115] and Atabay et al. [114], offer greater flexibility in detecting a wider range of behaviors through data analysis. While these methods may require more advanced data processing and model training for optimal accuracy, they are often more scalable and less reliant on specific hardware setups. The summary of hardware devices used to develop the IoT-based cheating detection system is shown in Table 15.

Table 15 Summary of hardware utilization in studies on automated cheating detection using IoT and emerging technologies

Author	Webcam/camera	Microphone/speaker	Mouse/Cikstream	Keystrokes	Other devices
Shevale et al. [109]	✓	✓	×	×	Standard Internet Connectivity
Ho et al. [110]	×	×	×	×	Raspberry Pi Zero W
Nguyen et al. [111]	✓	×	×	×	NVIDIA Jetson Nano, Dual-camera setup
La Roca et al. [112]	×	×	×	×	×
Plochaet et al. [113]	✓	✓	×	×	Fisheye Lens
Atabay et al. [114]	✓	×	×	×	×
Bommireddy et al. [115]	✓	✓	×	×	×

Table 16 Comparative analysis of automated proctoring systems using IoT and emerging technologies

Author	Year	Characteristics	Limitations	Proposed model	Performance	Datasets
Tweissi et al. [30]	2021	Focuses on IoT for real-time monitoring, Discusses ethical, technical, and legal challenges	No actual implementation, No empirical validation of the architecture	Conceptual IoT-based proctoring framework	Not mentioned	Own custom-made
Nguyen et al. [111]	2022	AI + IoT for real-time cheating detection; High detection accuracy	May fail under poor lighting or poor preparation	Dual-camera with AI-based fraud detection	85%+ accuracy	Video from online exams
Sangeeta Gulia et al. [116]	2022	Uses image processing for face tracking and eye movement	Only simulated data; lacks real-world validation; Not tested on diverse environments	AI-based automated proctoring using webcam	Reported qualitatively	Simulated environment, webcam feed
De La Roca et al. [112]	2022	Improved integrity/fairness perceptions	Stress and access issues for students	Survey and exam evaluation in MOOCs	High integrity satisfaction	Survey from MOOCs
Plochaet et al. [113]	2022	Combines anomaly, voice, and face detection	High false positives; Needs more training	Unified multi-modal detection system	TPR: 0.78, FPR: 0.19	Actor-based exam videos
Atabay et al. [114]	2022	GAN-based motion anomaly detection (skeleton)	Limited event detection accuracy	BeatGAN on bone-length skeleton data	Good anomaly score on motion	Skeleton from exam videos
Shevale et al. [109]	2023	Voice navigation for disabled users; scalable system	Privacy and noise-related issues not addressed	MERN + Python speech + client-server design	96.6% word accuracy (no full metric)	Simulated interaction dataset
Ho et al. [110]	2023	Lightweight IoT proctoring; no install needed	Works only on Windows	RAPID: Pi + Flask + PowerShell system	Effective for system-level monitoring	Local logs and traffic
Bom-mireddy et al. [115]	2023	Uses pose, voice, object detection for cheating	Limited behavior coverage	Multi-modal AI detection with real deployment	Effective in exams	Audio-video from real settings
Zuo et al. [117]	2024	Real-time cheating detection; custom dataset for paper-based exams	Limited contextual understanding in base YOLOv8, Attention modules may sometimes degrade performance	YOLOv8 + ECA attention mechanism	Accuracy: 82.71%, Subclass accuracy: $\geq$ 84.13%, FPS: 27	Custom-built dataset; 72,000 frames from two 20-min videos filmed in real classrooms

3.8 AI-enhanced learning and integrity

Recent studies have further expanded the role of AI-based technologies in education. [118] investigated the intersection of AI and sustainability in higher education, demonstrating how ethical AI frameworks can enhance operational efficiency while reducing environmental impact. [119] provide a comprehensive review of Large Language Models' (LLMs) applications in digital education, emphasizing both, their capacity to personalize learning, such as through tailored explanations, and critical issues including data privacy (58% of surveyed institutions reported concern) and algorithmic bias.

[120] show that Virtual Reality (VR) systems grounded in educational theory increase engineering students' retention scores by an average of 23%, also facilitating embodied cognition and situational engagement. Additionally, [121] demonstrate that coupling activity-based learning with ChatGPT results in a 30% rise in classroom engagement and a 0.5 standard deviation improvement in critical-thinking assessments. These findings highlight how LLMs, ChatGPT, VR, and educational theories can be effectively leveraged to enhance learner outcomes and uphold academic integrity. Also, a systematic review [122] highlighted the implementation of LLMs in higher education, noting their strengths in automated tutoring and content generation, alongside challenges in privacy and academic integrity. These recent contributions underscore the evolving landscape of AI-enhanced education and reinforce the relevance of our focus on ML, DL, and IoT in proctoring systems.

4 Major findings

This section presents a rigorous evaluation of the findings from our extensive literature review and model performance comparisons across CNN, RNN, LSTM, k-NN, and hybrid AI models applied to AIPS. We organize the discussion into the following key subsections: performance of DL models, comparative analysis of ML approaches, the promise of hybrid models, and critical reflections on limitations and future advancements.

4.1 Datasets used in AI-based proctoring research

The performance of AI-based proctoring systems is strongly influenced by the datasets used during development and evaluation. High-quality datasets enhance generalization, fairness, and robustness of detection mechanisms. In this section, we discuss publicly available datasets relevant to proctoring and describe custom datasets created by individual researchers for experimental purposes. Despite increasing research in automated proctoring, standardized publicly available datasets remain limited. The majority of existing datasets are either specific to a particular domain, or were collected for a research project as opposed to an educational setting. Publicly available datasets are often hard to acquire due to privacy, consent, and institutional restrictions. This leads to challenges in replication and evaluating performance relative to previous work. Even when publicly available, such as the Cohn Kanade Facial Expression Data Set (CK)/CK+ or LFW datasets, researchers still often would need to preprocess the datasets or adapt them for the proctoring use cases. Furthermore, custom datasets may have the benefit of containing richer multimodal or context specific dimensions, however they are typically not available for use in the wider research community. However, several datasets from related domains such as face recognition, eye-gaze detection, and authorship analysis have been utilized:

- *CK/CK+*: These datasets, widely used in facial expression recognition research, include sequences of facial images labeled with discrete emotional states. Researchers such as Mohamad et al. [69] leveraged CK/CK+ datasets to train CNN models for behavior detection during exams, particularly facial emotion shifts that may signal stress, distraction, or cheating cues. However, these datasets were originally not developed for proctoring purposes, which limits their contextual relevance.

Table 17 Datasets used in AI-based proctoring research with availability status

Dataset	Description/use in AIPS research	Limitations	Availability
CK / CK+	Extensively utilized in research investigating facial expression recognition; represents time-ordered sequences of facial images accompanied with labeling of a discrete emotional state; Mohamad et al. [69] have also used this dataset to train CNN models for behavior detection during exams based on the recognition of facial emotion transitions, indicating appearing stress or distractions	Originally not designed for exam proctoring; little context-specific relevance	Partially accessible to the public (requires a non-commercial research license and user agreement)
LFW	Video dataset primarily employed for face verification; establishes disaggregate expectations on face recognition performance from unconstrained environments; Guney et al. [65] employed the dataset to train the facial identification modules of vProctor	Limited to relevance of exam context; not multimodal	Available to the public for research, but there may be commercial restrictions
COCO Dataset	Wide set of labeled images for the purposes of object detection and localization; used to detect prohibited objects being used in the student environment (e.g., mobile phone or notes); hybrid proctoring model [86]	Not for academic contexts; no contextual exam data	Accessible to the public to use for research, under a standard license
EXARA-2015 Corpus	Compiled as part of the first International Competition on Arabic Plagiarism Detection; includes various forms of plagiarism; Nagoudi et al. [54] used the corpus to train ML classifiers based on text similarity and text structure manipulation	Supports textual cheating detection; no multimodal behavior data to make a full AIPS determination	Available to the public (Zenodo repository)
CEN	Over 25 million words across 292 novels produced by 25 authors; corpus used for stylometric studies and authorship profiling [55] and also applicable for intrinsic plagiarism application	Domain gap to student academic writing; not captured in exam contexts	Unclear / partially available to the public; licensing and access are not fully documented

- *LFW*: Used primarily for face verification, the LFW dataset provides a benchmark for face recognition performance in unconstrained environments. Studies such as Guney et al. [65] utilize LFW to train facial identification modules in their vProctor system.
- *COCO Dataset*: Although not tailored to academic settings, the COCO dataset offers a broad array of labeled images for object detection and localization tasks. It has been employed to detect unauthorized objects in students' environments (e.g., mobile phones, notes) in hybrid proctoring models [86].
- *EXARA-2015 Corpus*: Developed during the 1st International Competition on Arabic Plagiarism Detection, this dataset includes multiple plagiarism types and was used by Nagoudi et al. [54] to train ML classifiers on text similarity and structure manipulation. While it supports textual cheating detection, it lacks multimodal behavioral data necessary for full AIPS evaluation.
- *CEN*: With over 25 million words across 292 novels by 25 authors, CEN has been widely adopted for stylometric and authorship profiling studies [55]. It is particularly useful for intrinsic plagiarism detection, though it also suffers from a domain gap relative to actual academic writing by students.

Despite the contributions of these datasets (Table 17), they often fall short in three critical areas for AIPS research: firstly, contextual relevance, as many datasets were not recorded in actual examination environments, limiting their applicability; secondly, multimodal synchronization, due to the absence of aligned data streams such as audio,

video, and behavioral logs that are essential for holistic analysis; and thirdly, ground truth labels, as most datasets lack proper annotations distinguishing cheating from non-cheating behavior, which is vital for training and evaluation. To address the limitations of existing public datasets, many researchers have developed custom datasets tailored to online or offline proctoring environments, offering real-world behavioral data despite often being unavailable publicly due to privacy concerns. Jalali et al. [20] created a dataset with 50 reference and 10 test images per student under both normal and cheating conditions to detect seat vacancy and presence anomalies. Chuang et al. [50] captured webcam footage from 42 students during test sessions, logging head pose and timing variations for LR analysis. Sangalli et al. [57] analyzed interaction logs from the online course “Jugando con Android” to uncover coordinated cheating through co-occurrence and identity mismatches. Soma et al. used images from two institutions to detect hall ticket forgeries for ANN training. Borse et al. [79] simulated various cheating behaviors in a custom video dataset to train 3D CNNs using optical flow. Ferdosi et al. [70] annotated behavioral changes in pen-and-paper exam recordings, labeling both cheating and non-cheating instances. Yaqub et al. [61] developed a privacy-preserving dataset using blurred exam videos to detect motion anomalies. Lastly, Nabil et al. [66] collected keystroke and mouse data from a microprocessor exam, enabling a RF model to detect cheating with over 91% accuracy.

In conclusion, the lack of standardized, openly available proctoring datasets continues to be a significant barrier to replicable research. Datasets that are anonymized, standardized, and opened to public use would greatly assist in enhancing the comparability, reproducibility, and generalizability of AI-based proctoring systems. Researchers should also report dataset specificities (e.g., characteristics, limitations in access, details about ethics) when they report experiments.

4.2 Performance of convolutional neural networks in automated proctoring systems

CNNs have emerged as foundational tools in visual-based proctoring tasks due to their ability to extract and process spatial features from images and video streams. Numerous studies report their effectiveness in facial recognition, gesture analysis, and unauthorized object detection during online assessments. Hybrid CNN frameworks have shown particularly promising outcomes. For instance, CNN-RNN combinations achieved up to 99% accuracy, CNN with ResNet-34 attained 99.38%, and CNN integrated with Gaussian-based Discrete Fourier Transform (DFT) reached an impressive 99.8%. These architectures are optimized for capturing subtle facial cues and motion anomalies. Several models in prior studies report unusually high accuracy (up to 100%), such as in [20] and [70]. However, these results often come from small, in-house datasets with limited diversity. Such outcomes must be interpreted cautiously, as they may indicate overfitting or lack of cross-validation. Metrics such as precision, recall, and F1-score provide a more robust picture of model performance and should accompany accuracy in future evaluations. In contrast, simpler CNNs such as DesNet (91.4%), GoogleNet (91.2%), and ResNet (90.7%) performed slightly lower, indicating that architectural complexity and contextual modeling play a critical role in system accuracy. Notably, AlexNet and OpenPose models achieved only 77.8%, highlighting limitations in handling dynamic, real-time tasks suggesting the need for deeper and more specialized CNN variants in this domain.

4.3 Critical evaluation of reported results

While many reviewed models achieve high accuracy rates, their credibility varies depending on dataset size, labeling quality, and validation strategies. For instance, the 100% accuracy reported by Jalali et al. [20] is derived from only 50 reference images in controlled conditions, which may not reflect real-world diversity. Similarly, Ferdosi et al. [70] report near-perfect performance in non-cheating cases but do not fully address edge cases such as occlusion or student movement. Moreover, DL models typically require large and diverse datasets to generalize well. Without sufficient data augmentation or stratified sampling, the risk of overfitting remains high. It is also crucial to examine class imbalance, particularly in cheating vs. non-cheating classes, as accuracy alone can mask poor recall in minority categories.

4.4 Machine learning models and their role in behavioral analysis

In addition to image-based systems, traditional ML algorithms such as k-NN, SVM, and RF have been utilized to analyze behavioral data derived from keystrokes, browser activity, and biometric logs. k-NN classifiers combined with MediaPipe achieved up to 100% accuracy in controlled environments, showcasing their potential for fast deployment in small-scale systems. SVM models, on the other hand, demonstrated 95% accuracy and were widely praised for their robustness in supervised learning contexts. Despite these strengths, simpler models such as the Eigenface method with SVM struggled in practical deployment, achieving only 61% accuracy due to sensitivity to environmental noise and poor generalization under different lighting conditions.

4.5 Sequential modeling architectures

RNN and LSTM networks have demonstrated superior capability in modeling temporal behaviors, such as gaze transitions, head tilts, and gesture sequences over time. DenseLSTM models achieved 90.7% accuracy, while ConvLSTM with Long-term Recurrent Convolutional Networks (LRCN) reached 100%. In order to determine the efficacy of both recurrent architectures we conducted a systematic review of previous studies. The baseline RNN achieved an accuracy of 78.9% with the reduction in performance mainly attributed to its difficulties in capturing long-term temporal dependencies and vanishing gradients. The LSTM on the other hand achieved a higher accuracy and displayed a more stable learning behavior.

4.6 Hybrid AI architectures for real-time cheating detection

The synergy between CNNs (for spatial feature extraction) and RNNs or LSTMs (for temporal modeling) suggests that hybrid architectures may offer the most promising route for real-time, scalable, and accurate AIPS. These models are capable of simultaneously recognizing instantaneous facial or object-level anomalies while analyzing behavioral trends across time windows. Further enhancement can be achieved by integrating IoT devices and biometric sensors (e.g., EEG, eye-trackers, fingerprint scanners) to extend the sensing capability beyond visual analysis. However, this requires addressing interoperability, real-time latency, and privacy concerns.

4.7 Critical discussion of findings, limitations, and future directions of AIPS

AIPS have demonstrated significant potential in addressing the challenges of maintaining academic integrity in remote and hybrid learning environments. The findings from recent studies highlight the strengths of AIPS in detecting diverse cheating behaviors through facial recognition, eye tracking, audio analysis, and behavioral biometrics using ML and DL algorithms. These systems reduce human dependency and improve scalability, allowing institutions to monitor large-scale online assessments with minimal logistical burden. However, despite these advancements, AIPS present several limitations. First, the high false-positive and false-negative rates in some systems raise concerns about the reliability and fairness of decision-making, especially for students from diverse cultural and physical backgrounds. Second, the dependence on webcams, microphones, and high-speed internet excludes students in rural or under-resourced regions, thereby creating a digital divide. Third, many AIPS models are trained on biased or narrow datasets, making them less generalizable in real-world environments. Fourth, privacy and ethical concerns persist, as continuous surveillance may intrude on personal space and violate regulatory frameworks such as GDPR and the PDP Act. Looking ahead, the future of AIPS lies in developing hybrid models that combine visual, audio, and behavioral analytics with privacy-preserving techniques, such as on-device processing and anonymized data handling. Researchers must also focus on creating explainable AI models to increase trust among stakeholders. Additionally, the integration of emerging technologies, such as VR for immersive assessments and LLMs for real-time feedback, could transform AIPS into more adaptive, student-centered systems. Standardized benchmarks and large-scale validation across varied populations are crucial to ensuring the robustness and fairness of these systems in educational settings. When comparing ML and DL approaches, ML models such as SVM and RF exhibit better interpretability and require less training data, making them more suitable for structured inputs like keystroke dynamics or exam logs. In contrast, DL models, especially CNN-based systems, outperform ML techniques in visual analysis tasks like gaze detection and face recognition due to their hierarchical feature extraction capabilities. However, DL models often require large annotated datasets and substantial computational resources. ML methods may fail to capture complex patterns in visual data, while DL systems may overfit or misclassify in scenarios with limited training samples or varied lighting conditions.

5 Future directions, issues, and challenge

In light of our comprehensive review of 80 studies, several persistent limitations and emerging opportunities have been identified across ML, DL, and IoT-enabled proctoring systems. This section highlights the most pressing challenges and outlines potential directions for future research and development.

5.1 Technical challenges in AI-based proctoring models

DL models such as CNNs, LSTMs, and RNNs demonstrate strong potential for behavior recognition, yet face critical limitations due to their reliance on large, annotated datasets for effective training and adversarial attacks in AI models which can be described as slight alterations of inputs that would trick an AI algorithm [123]. The scarcity of high-quality, labeled exam monitoring data hampers generalization across diverse testing conditions. Moreover, DL systems are computationally intensive and require high-end

Graphics Processing Units (GPUs) or cloud-based infrastructure, which may not be viable in resource-constrained or remote environments. Overfitting is another concern when training on small or biased datasets, potentially leading to unreliable or unfair detection outcomes. These limitations become even more prominent in the proctoring context since students can exhibit very different behaviours in different environments, from different camera angles and lighting conditions and on different devices, which adds further noise that is not reliably managed by models trained on small data. Classical ML techniques, including SVM, DT, and RF, offer better interpretability in some contexts, yet they often lack robustness against adversarial examples and perform poorly on high-dimensional visual data. Furthermore, the "black-box" nature of many ML and DL models makes it difficult for educators to interpret decisions, which can erode trust in automated assessment systems. Lack of robustness becomes an issue in traditional testing scenarios where students inadvertently trigger false alerts from natural behaviours or momentary face occlusions or due to lack of quality or resolution of webcams, for example. The variable conditions erode the trustworthiness of the decision made by automated algorithms.

5.2 Fairness, bias, and interpretability issues

Bias in AI models remains a central ethical and technical concern. Studies have shown that facial detection and gaze estimation models may exhibit discriminatory behavior based on race, gender, or lighting conditions, leading to inequitable outcomes for marginalized groups. To address these disparities, future research must focus on building fairer systems through the use of balanced and demographically diverse datasets, as well as bias-mitigation techniques such as reweighting, adversarial de-biasing, and interpretable fairness-aware learning. In a proctoring context, these biases directly jeopardize the ability to manage some students as trustworthy or to risk misclassifying certain students as suspicious. For example, students of colour—usually with darker skin tones—are more likely to be misclassified by detection algorithms, students with certain neurodivergent behaviours are more likely to trigger the anomaly alerts, and gaze behaviors that differ with different cultural practices may be classified as cheating behaviors. Additionally, interpretability remains underexplored in proctoring systems. Model explainability is essential for validating system decisions, particularly when students are penalized based on automated judgments. Techniques such as SHAP, LIME, Grad-CAM, and attention visualization can help demystify model behavior, allowing educators to assess the rationale behind alerts and flagging. Without necessary interpretability, students may face discipline as an outcome of their suspicious misclassifications due to the detection algorithms, and this threatens academic fairness and means that fairness in the context of the high-stakes assessment environment becomes muddled.

5.3 Privacy, security, and integration challenges

The integration of IoT devices into proctoring frameworks introduces several risks and engineering challenges. Systems that collect biometric and behavioral data (e.g., webcam feeds, keystroke dynamics, or audio signals) must comply with global privacy regulations such as GDPR and institutional data use policies. Secure data transmission, storage, and access control are vital to protect students from privacy breaches. In online exams and proctoring scenarios, the privacy risks of highly sensitive recordings of people's footage,

recorded in their own homes and study spaces, and private spaces are more significant than when in traditional IoT context. Moreover, technical integration of sensors, edge devices, and cloud-based analytics requires standardization across protocols. Latency in real-time processing—especially in cloud-based deployments—can cause detection delays, reducing the system's responsiveness and reliability. Future work should explore lightweight edge-computing approaches and privacy-preserving techniques such as federated learning or homomorphic encryption. In a proctoring workflow, any slight amount of latency will mean some events are missed altogether, any warnings will be presented late, or some signaling can be sent, but the multi-modality of audio and video may lag at different times, meaning the workflow causes an exchange of reliability for exam monitors.

5.4 Human-centric issues

The future of proctoring lies in systems that not only secure academic integrity but also respect the psychological well-being and comfort of students. Traditional camera-based surveillance has been associated with increased stress and test anxiety, which can adversely affect student performance. Less intrusive approaches, such as anomaly detection based on typing cadence, mouse movement, or passive engagement metrics, offer promising alternatives. These human-centred approaches are especially relevant, because while excessive monitoring will cause students to change their natural behaviour, this ironically makes it more likely, automated systems detect behaviours to flag as suspicious. While TensorFlow and PyTorch dominate DL implementation, their use varies depending on model complexity, hardware support, and community adoption. PyTorch offers more flexibility for research-level experimentation, whereas TensorFlow is preferred in production environments. OpenCV, widely used for image pre-processing and traditional computer vision tasks, integrates effectively with both platforms. However, its use in complex DL pipelines is limited. They also engage in the important model-selection trade-offs inherent in proctoring applications, because institutions differ so much from reliable compute capacity all the way to the quality of the students' devices and types of examination format. ML methods such as SVM and DT remain relevant, especially in constrained environments with smaller datasets or text-based input features. For example, plagiarism detection models using stylometric analysis or word embeddings are more efficiently handled with lightweight classifiers. The choice between classical ML and DL should be justified by data modality, model interpretability, and computational resources. In addition, adaptive proctoring systems capable of dynamically adjusting monitoring intensity based on real-time student behavior or risk assessment can reduce false positives and improve user experience. These systems can be context-aware and seamlessly integrate with Learning Management Systems (LMS) and virtual classrooms to offer a holistic, scalable, and intelligent proctoring ecosystem. These adaptive systems can identify which behaviours are more indicative, adjudicate between the audio and video streams, and set thresholds with sensitivity into "deterrent" ranges to match the context with more effective outcomes regarding proctoring outcomes or performance as it relates to fairness.

6 Discussion

This study presents a structured analysis of AIPS by categorizing and comparing various ML, DL, IoT, and hybrid approaches. The review reveals that while CNNs, LSTMs, and hybrid DL architectures have shown high accuracy in visual behavior monitoring, their reliance on large datasets and compute resources limits scalability across institutions with low resources. Meanwhile, simpler models like DTs and heuristic systems offer transparency and ease of deployment but struggle with generalization and adaptability to diverse cheating patterns. IoT-based models provide real-time, non-visual data streams (e.g., environmental sensors, keystroke logging) but raise privacy concerns and integration overheads. Additionally, behavioral biometrics (e.g., keystroke dynamics, gaze tracking) represent promising privacy-preserving alternatives but suffer from performance degradation under noise and cross-user variability. It is also evident that most existing approaches evaluate accuracy in isolation, without balancing trade-offs between interpretability, robustness, fairness, and ethical feasibility. Additionally, forthcoming AI-staffed proctoring systems should explicitly outline the general identities, the sources, and some expected uses of common components to ensure appropriate, credible, and ethically justified implementations. Furthermore, few models integrate multimodal behavioral signals effectively missing opportunities for more robust and context-aware detection. The deployment of AI-driven proctoring systems raises several ethical concerns. Algorithmic surveillance during exams may induce psychological stress and alter student behavior. Additionally, many proctoring systems rely on facial recognition, gaze tracking, and biometric data, posing significant privacy risks. Without clear informed consent policies, institutions may unintentionally violate user rights. Moreover, the storage and processing of biometric information require robust data protection strategies to prevent misuse or breaches. Addressing these ethical issues is essential for the sustainable adoption of AI proctoring technologies. Our findings suggest that while current methods are technically powerful, they are often siloed. A future shift towards ethical, scalable, and context-sensitive hybrid systems—integrating visual, behavioral, and environmental cues—is necessary to build practical AIPS for global educational settings.

7 Conclusion

This study presents a comprehensive review of AIPS by synthesizing 80 peer-reviewed studies from 2014 to 2024, spanning ML, DL, IoT-based, and hybrid approaches. As educational institutions worldwide, specifically higher education institutions increasingly shift toward digital assessment platforms, maintaining academic integrity has become a critical challenge. The research identifies the significant role of AI in detecting and preventing dishonest behaviors during online examinations through automated monitoring of facial expressions, gaze direction, keystrokes, audio cues, and environmental interactions. The findings reveal that models such as CNNs, RNNs, SVMs, and RFs show promising results in real-time cheating detection. IoT integration further enhances system responsiveness and coverage through biometric verification, gaze tracking, and posture estimation. Despite these advancements, the study highlights recurring challenges, including false positives, dataset biases, lack of contextual awareness, and ethical concerns around user privacy. To address these issues, the paper advocates for the development of unified, multimodal, and privacy-preserving AIPS frameworks that leverage edge computing, secure data handling, and diverse behavioral datasets. The proposed

conceptual architecture integrates biometric authentication, behavior analysis, and environmental monitoring for a robust and scalable solution. Despite the comprehensive scope of this review, several limitations must be acknowledged. First, some included studies rely on small-scale or simulated datasets, limiting real-world generalizability. Second, the rapid evolution of AI methods means that newer or hybrid frameworks may emerge that surpass the performance of those reviewed. Lastly, ethical evaluations remain underexplored in the literature, particularly regarding long-term data storage and user consent enforcement. Future research should prioritize the development of lightweight, privacy-preserving AI models that can operate on edge devices for scalable deployment. There is also a pressing need for longitudinal studies evaluating the psychological and pedagogical impact of algorithmic proctoring. Furthermore, interdisciplinary collaborations involving AI researchers, ethicists, educators, and policymakers are crucial to ensure that these systems remain both effective and equitable.

Acknowledgements

The authors would like to thank all anonymous reviewers for their valuable comments and suggestions, which greatly improved the quality of this paper. I express my gratitude to Panjab University for providing an intellectually enriching environment and supporting research endeavors. Appreciation is also extended to faculty from the university for their valuable support throughout the research process.

Author contributions

M.M. and I.C. jointly conceived the research idea and developed the methodology for the study. M.M. contributed to conceptualization, formal analysis, methodology, data curation, and writing the original draft. I.C. took the responsibility of review, editing, supervision, and project administration.

Funding

The authors did not receive any specific funding for this research.

Data availability

Not applicable.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing of interests

The authors declare that they have no known competing financial interests or personal relationships that could have influenced this work.

Received: 2 September 2025 / Accepted: 5 February 2026

Published online: 13 February 2026

References

1. Muzaffar AW, Tahir M, Anwar MW, Chaudry Q, Mir SR, Rasheed Y. A systematic review of online exams solutions in e-learning: techniques, tools, and global adoption. *IEEE Access*. 2021;9:32689–712.
2. Zhou T, Jiao H. Exploration of the stacking ensemble machine learning algorithm for cheating detection in large-scale assessment. *Educ Psychol Measur*. 2023;83(4):831–54.
3. Potluri T, Sistla VPK. A comprehensive survey on the AI based fully automated online proctoring systems to detect anomalous behavior of the examinee. In: 2022 International conference on recent trends in microelectronics, automation, computing and communications systems (ICMAACC), pp. 407–411, IEEE, 2022.
4. Blau I, Eshet-Alkalai Y. The ethical dissonance in digital and non-digital learning environments: Does technology promotes cheating among middle school students? *Comput Hum Behav*. 2017;73:629–37.
5. Tuah NAA, Naing L. Is online assessment in higher education institutions during covid-19 pandemic reliable? *Siriraj Med J*. 2021;73(1):61–8.
6. von Jena ZA. The cognitive conditions associated with academic dishonesty in university students and its effect on society. *UC Merced Undergraduate Res J*. 2020;1.
7. Alessio HM, Malay N, Maurer K, Bailer AJ, Rubin B. Examining the effect of proctoring on online test scores. *Online Learn*. 2017;21(1):146–61.
8. Newton PM. How common is commercial contract cheating in higher education and is it increasing? A systematic review. In: *Frontiers in Education, Frontiers Media SA*, 2018;3:67.

9. Brimble M. Why students cheat: an exploration of the motivators of student academic dishonesty in higher education. *Int J Educ Integr.* 2016;12(1):1–14.
10. Marchant GJ, Baek J. The role of stress, anxiety, and emotions in academic misconduct. *J Acad Ethics.* 2021;19(3):295–310.
11. Elsalem L, Al-Azzam N, Jum'ah AA. AI-driven proctoring systems in remote exams: advances, limitations, and future directions. *Comput Educ Artif Intell.* 2023;4:100133.
12. Al-Fraihat D, Joy M, Sinclair J. Ethical and practical implications of AI in remote proctoring. *J Educ Comput Res.* 2022;60(2):376–98.
13. Stike T. Online proctoring services compared | 5-part guide. <https://honorlock.com/blog/definitive-guide-for-comparing-online-proctoring-services/>, 2021. Accessed: 5 January, 2024.
14. Kusuma GP, Suhartono D. Artificial intelligence in academic proctoring: a systematic review. *Proc Comput Sci.* 2022;197:512–9.
15. Rosenberg J, Chopra S. Ai-driven academic integrity: redefining exam surveillance. *AI & Educ J.* 2020;14(2):75–88.
16. Saba T, Rehman A, Jamail NSM, Marie-Sainte SL, Raza M, Sharif M. Categorizing the students' activities for automated exam proctoring using proposed deep l2-graftnet cnn network and aso based feature selection approach. *IEEE Access.* 2021;9:47639–56.
17. Hendryli J, Fanany MI. Classifying abnormal activities in exam using multi-class markov chain lda based on modec features. In: 2016 4th international conference on information and communication technology (ICICT), pp. 1–6, IEEE, 2016.
18. Haytom M, Rosenberger C, Charrier C, Zhu C, Régner C. Identity verification and fraud detection during online exams with a privacy compliant biometric system. *ICETE.* 2020;2:451–8.
19. Asadullah M, Nisar S. An automated technique for cheating detection. In: 2016 sixth international conference on innovative computing technology (INTECH), IEEE, 2016. pp. 251–255.
20. Jalali K, Noorbehhahani F. An automatic method for cheating detection in online exams by processing the student's webcam images; in 3rd conference on electrical and computer engineering technology (E-Tech 2017), Tehran, Iran, 2017. pp. 1–6.
21. Varun R, Rao PP, Manohar S, Nagsharan A, et al. Virtual overseer for examination using artificial intelligence. *Indonesian J Inform Syst.* 2024;6(2):174–82.
22. Akour I, Alshurideh M, Al Kurdi B, Al Ali A, Salloom S, et al. Using machine learning algorithms to predict people's intention to use mobile learning platforms during the covid-19 pandemic: Machine learning approach. *JMIR Med Educ.* 2021;7(1):e24032.
23. Kaddoura S, Gumaei A. Towards effective and efficient online exam systems using deep learning-based cheating detection approach. *Intell Syst Appl.* 2022;16:200153.
24. Malhotra N, Suri R, Verma P, Kumar R. Smart artificial intelligence based online proctoring system. In: 2022 IEEE Delhi section conference (DELCON), IEEE, 2022. pp. 1–5.
25. Soltane M, Laouar MR. A smart system to detect cheating in the online exam. In: 2021 International conference on information systems and advanced technologies (ICISAT), IEEE, 2021. pp. 1–5.
26. Tiong LCO, Lee HJ, Lim KL. Online assessment misconduct detection using internet protocol and behavioural classification. 2022. arXiv preprint [arXiv:2201.13226](https://arxiv.org/abs/2201.13226).
27. Oravec JA. Ai, biometric analysis, and emerging cheating detection systems: The engineering of academic integrity? *Educ Policy Anal Archives.* 2022;30(175):n175.
28. Nigam A, Pasricha R, Singh T, Churi P. A systematic review on ai-based proctoring systems: past, present and future. *Educ Inf Technol.* 2021;26(5):6421–45.
29. Han S, Nikou S, Yilma AW. Digital proctoring in higher education: a systematic literature review. *Int J Educ Manag.* 2024;38(1):265–85.
30. Tweissi A, Al Etaiwi W, Al ED. The accuracy of ai-based automatic proctoring in online exams. *Electron J e-Learn.* 2022;20(4):419–35.
31. Nafrees ACM, Bharathi B, Sasubilli D, Komma SSR. Machine learning technique for facial datasets to detect examination fraudulent activities in the online examination: a systematic review approach. In: 2022 IEEE international conference on current development in engineering and technology (CCET), IEEE, 2022. pp. 1–6.
32. Shrivastava AK, Sonawane SD. Real-time student monitoring using deep learning. *Proc Comput Sci.* 2022;203:210–7.
33. Nwosu E, et al. Evaluating bias and accuracy in ai-powered proctoring tools. *IEEE Access.* 2023;11:88742–56.
34. Rana V, Garg S. Challenges in generalizing ai-based proctoring systems across geographies. *Int J Educ Technol.* 2021;18(3):131–45.
35. Miranda R, et al. Ethical challenges in automated surveillance for educational contexts. *Comput Educ.* 2023;200:104743.
36. Alshammari T, Ahmed M. Equity in remote assessment: tech barriers and ai implications. *Educ Inf Technol.* 2022;27(6):8771–90.
37. Ruhland A, Singh P. From anomaly detection to contextual ai: the future of proctoring. *ACM Transactions on Learning Technologies*, in press. 2024.
38. Jadhav R, Mehta A. Gaze tracking for intelligent proctoring: Deep learning advances. *Pattern Recogn Lett.* 2023;167:123–30.
39. Liu Z, et al. Multicultural dataset construction for fair ai in education. *Comput Human Behav Rep.* 2024;10:100155.
40. Fang L, Banerjee R. Privacy-preserving ai for online assessment monitoring. *J Educ Comput Res.* 2021;59(8):1427–44.
41. Kumar R, Bansal N. Multimodal fusion for proctoring: towards smarter surveillance. *IEEE Trans Affect Comput.* 2023;14(1):115–26.
42. Raj RV, Narayanan SA, Bijlani K. Heuristic-based automatic online proctoring system. In: 2015 IEEE 15th international conference on advanced learning technologies, IEEE, 2015. pp. 458–459.
43. Li X, Chang K-M, Yuan Y, Hauptmann A. Massive open online proctor: Protecting the credibility of moocs certificates. In: Proceedings of the 18th ACM conference on computer supported cooperative work & social computing, 2015. pp. 1129–1137.
44. Soma S, Vidyashree S. An automated fraud detection of hall ticket in an offline examination system using ann classifier. *Int J Comput Appl.* 2015;126(8).
45. Abdallah A, Maarof MA, Zainal A. Fraud detection system: a survey. *J Netw Comput Appl.* 2016;68:90–113.

46. AlSallal M, Iqbal R, Amin S, James A, Palade V. An integrated machine learning approach for extrinsic plagiarism detection. In: 2016 9th international conference on developments in eSystems engineering (DeSE), IEEE, 2016. pp. 203–208.
47. Cote M, Jean F, Albu AB, Capson D. Video summarization for remote invigilation of online exams. In: 2016 IEEE winter conference on applications of computer vision (WACV), IEEE, 2016. pp. 1–9.
48. Staibitz T, Teusner R, Renz J, Meinel C. An experiment in automated proctoring. *Res Track*, 2016;41.
49. Northcutt CG. Classification with noisy labels: "Multiple Account" cheating detection in Open Online Courses. PhD thesis, Massachusetts Institute of Technology, 2017.
50. Chuang CY, Craig SD, Femiani J. Detecting probable cheating during online assessments based on time delay and head pose. *Higher Educ Res Develop*. 2017;36(6):1123–37.
51. Ruiperez-Valiente JA, Munoz-Merino PJ, Alexandron G, Pritchard DE. Using machine learning to detect 'multiple-account' cheating and analyze the influence of student and problem features. *IEEE Trans Learn Technol*. 2017;12(1):112–22.
52. Atoum Y, Chen L, Liu AX, Hsu SD, Liu X. Automated online exam proctoring. *IEEE Trans Multimedia*. 2017;19(7):1609–24.
53. Soman N, Devi MR, Srinivasa G. Detection of anomalous behavior in an examination hall towards automated proctoring. In: 2017 second international conference on electrical, computer and communication technologies (ICECCT), IEEE, 2017. pp. 1–6.
54. Cherroun H, Alshehri A, et al. Disguised plagiarism detection in arabic text documents. In: 2018 2nd international conference on natural language and speech processing (ICNLSP), IEEE, 2018. pp. 1–6.
55. AlSallal M, Iqbal R, Palade V, Amin S, Chang V. An integrated approach for intrinsic plagiarism detection. *Futur Gener Comput Syst*. 2019;96:700–12.
56. Qiubo H, Jingdong T, Guozheng F. Research on code plagiarism detection model based on random forest and gradient boosting decision tree. In: Proceedings of the 2019 international conference on data mining and machine learning, 2019. pp. 97–102.
57. Sangalli VA, Martinez-Muñoz G, Cañabate EP. Identifying cheating users in online courses. In: 2020 IEEE global engineering education conference (EDUCON), IEEE, 2020. pp. 1168–1175.
58. Ljubovic V, Pajic E. Plagiarism detection in computer programming using feature extraction from ultra-fine-grained repositories. *IEEE Access*. 2020;8:96505–14.
59. Sinha P, Yadav A, et al. Remote proctored theory and objective online examination. *Int J Adv Netw Appl*. 2020;11(6):4494–500.
60. Hoque MJ, Ahmed MR, Uddin MJ, Faisal MMA. Automation of traditional exam invigilation using cctv and bio-metric. *Int J Adv Comput Sci Appl*. 2020;11(6).
61. Yaqub W, Mohanty M, Suleiman B. Image-hashing-based anomaly detection for privacy-preserving online proctoring. 2021. arXiv preprint [arXiv:2107.09373](https://arxiv.org/abs/2107.09373).
62. Li H, Xu M, Wang Y, Wei H, Qu H. A visual analytics approach to facilitate the proctoring of online exams. In: Proceedings of the 2021 CHI conference on human factors in computing systems, 2021. pp. 1–17.
63. Geetha M, Latha R, Nivetha S, Hariprasath S, Gowtham S, Deepak C. Design of face detection and recognition system to monitor students during online examinations using machine learning algorithms. In: 2021 international conference on computer communication and informatics (ICCCI), IEEE, 2021. pp. 1–4.
64. Slusky L. Cybersecurity of online proctoring systems. *J Int Technol Inform Manag*. 2020;29(1):56–83.
65. Guney C, Akinci O, Çamoğlu K. Artificial learning-based proctoring solution for remote online assessments: "vproctor". *Int Arch Photogramm Remote Sens Spat Inf Sci*. 2021;46:235–8.
66. Nabil RH, Barid M, Sami A, Hossain MN, et al. An intelligent examination monitoring tool for online student evaluation. *Malaysian J Sci Adv Technol*. 2022;122–130.
67. Kasinathan V, Yan CE, Mustapha A, Hameed VA, Ching TH, Thiruchelvam V. Proctorex: an automated online exam proctoring system. *Math Stat Eng Appl*. 2022;71(32):876–89.
68. Garg M, Goel A. Preserving integrity in online assessment using feature engineering and machine learning. *Expert Syst Appl*. 2023;225:120111.
69. Mohamed Nafrees AC, Aysha Asra S, Kariapper R. Machine learning for autonomous online exam fraud detection: A concept design. *BOHR Int J Internet Things Artif Intell Mach Learn*. 2023.
70. Ferdosi B, Rahman M, Sakib A, Helaly T, et al. Modeling and classification of the behavioral patterns of students participating in online examination. *Human Behav Emerg Technol*. 2023;2023.
71. Desai R, Muni R, Madhani D, Joshi A. Predicting learning behavior using hyper-personalization and attentiveness measure. In: 2022 5th international conference on advances in science and technology (ICAST), IEEE, 2022. pp. 258–264.
72. Smirani LK, Boulahia JA. An algorithm based on convolutional neural networks to manage online exams via learning management system without using a webcam. *Int J Adv Comput Sci Appl*. 2022;13(3).
73. Breskina AA. Development of an automated online proctoring system. *Herald Adv Inf Technol*. 2023;6:163–73.
74. Hussein F, Al-Ahmad A, El-Salhi S, Alshdaifat E, Al-Hami M. Advances in contextual action recognition: automatic cheating detection using machine learning techniques. *Data*. 2022;7(9):122.
75. Cai X, Hu F, Ding L. Detecting abnormal behavior in examination surveillance video with 3d convolutional neural networks. In: 2016 6th international conference on digital home (ICDH), IEEE, 2016. pp. 20–24.
76. Hu S, Jia X, Fu Y. Research on abnormal behavior detection of online examination based on image information. In: 2018 10th international conference on intelligent human-machine systems and cybernetics (IHMSC), IEEE, 2018;vol. 2, pp. 88–91.
77. Asep HS, Bandung Y. A design of continuous user verification for online exam proctoring on m-learning. In: 2019 International conference on electrical engineering and informatics (ICEEI), IEEE, 2019. pp. 284–289.
78. Nishchal J, Reddy S, Navya PN. Automated cheating detection in exams using posture and emotion analysis. In: 2020 IEEE international conference on electronics, computing and communication technologies (CONECCT), IEEE, 2020. pp. 1–6.
79. Borse MR, Rath V, Pawar PG, Mirase PD, Paste YY. Detecting abnormal behavior in examination surveillance with 3d convolutional neural networks. In: 2016 6th International Conference on Digital Home (ICDH), IEEE, 2016. pp. 20–24.
80. Aini YK, Saleh A, Darwito HA. Implementation of background subtraction method to detect movement of examination participants. In: 2020 international electronics symposium (IES), IEEE, 2020. pp. 494–500.
81. Garg K, Verma K, Patidar K, Tejra N. Convolutional neural network based virtual exam controller. In: 2020 4th international conference on intelligent computing and control systems (ICICCS), IEEE, 2020. pp. 895–899.

82. Labayen M, Vea R, Flórez J, Aginako N, Sierra B. Online student authentication and proctoring system based on multimodal biometrics technology. *IEEE Access*. 2021;9:72398–411.
83. Jadi A. New detection cheating method of online-exams during covid-19 pandemic. *Int J Comput Sci Netw Secur*. 2021;21(4):123–30.
84. Ganidisastira AHS, Bandung Y. An incremental training on deep learning face recognition for m-learning online exam proctoring. In: 2021 IEEE Asia pacific conference on wireless and mobile (APWiMob), IEEE, 2021. pp. 213–219.
85. He K, Zhang X, Ren S, Sun J. Deep residual learning for image recognition. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016. pp. 770–778.
86. Lin T-Y, Maire M, Belongie S, Hays J, Perona P, Ramanan D, Dollár P, Zitnick CL. Microsoft coco: Common objects in context. In: *Computer Vision-ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6–12, 2014, Proceedings, Part V 13*, Springer, 2014. pp. 740–755.
87. Li B, Li E. Automated online proctoring system using gaze view tracking and custom object detection. In: 2022 IEEE MIT Undergraduate Research Technology Conference (URTC), IEEE, 2022. pp. 1–5.
88. Liu T. Ai proctoring for offline examinations with 2-longitudinal-stream convolutional neural networks. *Comput Educ Artif Intell*. 2023;4:100115.
89. Tejaswi P, Venkatramaphanikumar S, Kishore KVK. Proctor net: an AI framework for suspicious activity detection in online proctored examinations. *Measurement*. 2023;206:112266.
90. Potluri T, K VKK. An automated online proctoring system using attentive-net to assess student mischievous behavior. *Multimed Tools Appl*. 2023;1–30.
91. Yulita IN, Hariz FA, Suryana I, Prabuwo AS. Educational innovation faced with covid-19: deep learning for online exam cheating detection. *Educ Sci*. 2023;13(2):194.
92. Zaffar A, Jawad M, Shabbir M. A novel cnn-rnn model for e-cheating detection based on video surveillance. *Univ Wah J Comput Sci*. 2023;5:1–13.
93. Kulshrestha A, Gupta A, Singh U, Sharma A, Shukla A, Gautam R, Kumar P, Pandey D. Ai-based exam proctoring system. In: 2023 international conference on disruptive technologies (ICDT), IEEE, 2023. pp. 594–597.
94. Saravanan R. Automatic cheating detection in exam hall. *Authorea Preprints*, 2023.
95. Mohajer MM, Hassanpour H. Fast exam video summarization using targeted evaluation of scene changes based on user behavior. In: 2023 9th International conference on signal processing and intelligent systems (ICSPIS), IEEE, 2023. pp. 1–5.
96. Verma P, Malhotra N, Suri R, Kumar R. Automated smart artificial intelligence-based proctoring system using deep learning. *Soft Comput*. 2024;28(4):3479–89.
97. Anwar A. Difference between alexnet, vggnet, resnet and inception. <https://medium.com/data-science/the-w3h-of-alexnet-vggnet-resnet-and-inception-7baaaecccc96>, 2022;. Accessed: 5 January, 2024.
98. Staudemeyer RC, Morris ER. Understanding lstm-a tutorial into long short-term memory recurrent neural networks. 2019. arXiv preprint [arXiv:1909.09586](https://arxiv.org/abs/1909.09586).
99. Fan Y, Lu X, Li D, Liu Y. Video-based emotion recognition using cnn-rnn and c3d hybrid networks. In: *Proceedings of the 18th ACM international conference on multimodal interaction*, 2016. pp. 445–450.
100. Arinaldi A, Fanany MI. Cheating video description based on sequences of gestures. In: 2017 5th international conference on information and communication technology (IColC7), IEEE, 2017. pp. 1–6.
101. Kadyrov B, Kadyrov S, Makhmutova A. Automated reading detection in an online exam. *Int J Emerg Technol Learn*. 2022;17(22):4–19.
102. Tzeng J-W, Hsueh C-Y, Lee C-A, Shih W-Y. Identifying the correlation between online exam answer trajectory and test behavior based on artificial intelligence and eye movement detection technology. In: 2023 International Conference on Consumer Electronics-Taiwan (ICCE-Taiwan), IEEE, 2023. pp. 503–504.
103. Felsinger D, Halloluwa T, Fonseka I. Video based action detection for online exam proctoring in resource-constrained settings. *Educ Inform Technol*, 2023;1–15.
104. Tang S, Samuel S, Li Z. Detecting a typical test-taking behaviors with behavior prediction using lstm. *Psychol Test Assess Model*. 2023;65(2):76–124.
105. Nurpeisova A, Shaushenova A, Mutałova Z, Ongarbayeva M, Niyazbekova S, Bekenova A, et al. Research on the development of a proctoring system for conducting online exams in Kazakhstan. *Computation*. 2023;11(6):120.
106. Qi X, Liu C, Schuckers S. Iot edge device based key frame extraction for face in video recognition. In 2018th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGRID), IEEE, 2018. pp. 641–644.
107. Chitra R, Sundararajan M. "Iot-based remote exam proctoring system using deep learning," *J mbient Intell Humanized Comput*. 2022.
108. Ramesh K, Suresh M. An intelligent proctoring system using IoT and AI techniques. *Proc Comput Sci*. 2020;171:739–47.
109. Shevale P, Shitole V, More S, Gaykar Y, Gaigol A. Xampro (voice-based exam proctoring system) in 2023 11th International Conference on Emerging Trends in Engineering & Technology-Signal and Information Processing (ICETET-SIP), 2023. pp. 1–4, IEEE.
110. Ho JHY, Tan DZ, Yap JY, Tse KP, Abbas MFB, Loo AWY, Goh W. Iot-enhanced remote proctoring: A new paradigm for remote assessment integrity. In 2023 IEEE 35th international conference on software engineering education and training (CSEET&T), IEEE, 2023. p. 197–198.
111. Nguyen XH, Le-Pham VM, Than TT, Nguyen MS. Proctoring online exam using iot technology. In 2022 9th NAFOSTED conference on information and computer science (NICS), IEEE, 2022. pp. 7–12.
112. De La Roca M, Morales M, García-Cabot A. The impact of online proctoring on students' perception and level of satisfaction in a mooc. In 2022 IEEE learning with MOOCs (LWMOOCs), IEEE, 2022. pp. 23–27.
113. Plochaet J, Goedemé T. Towards automatic proctoring of online exams using video anomaly detection. *International Scientific Conferences on AI and Machine Learning*, 2022.
114. Atabay HA, Hassanpour H. Abnormal behavior detection in electronic-exam videos using beatgan. In 2022 8th Iranian conference on signal processing and intelligent systems (ICSPIS), IEEE, 2022. pp. 1–5.
115. Bommireddy LR, Marasu RT, Karanam RP, Sri KS. Smart proctoring system using ai. In 2023 3rd international conference on pervasive computing and social networking (ICPCSN), IEEE, 2023. pp. 591–593.
116. Lamba S, Sharma N. Deep learning-based multimodal cheating detection in online proctored exams. *J Electr Syst*, 2024;20(3).

117. Zuo Y, Chai SS, Goh KL. Cheating detection in examinations using improved yolov8 with attention mechanism. *J Comput Sci.* 2024;20:1668–80.
118. Khan S, Mazhar T, Shahzad T, Khan MA, Rehman AU, Saeed MM, et al. Harnessing ai for sustainable higher education: ethical considerations, operational efficiency, and future directions. *Discovery Sustain.* 2025;6:23.
119. Shahzad T, Mazhar T, Tariq MU, Ahmad W, Ouahada K, Hamam H. A comprehensive review of large language models: issues and solutions in learning environments. *Discovery Sustainability*, 2025.
120. Shah SFA, Mazhar T, Shahzad T, Khan MA, Ghadi YY, Hamam H. Integrating educational theories with virtual reality: enhancing engineering education and vr laboratories. *Soc Sci Humanities Open.* 2024;10:101207.
121. Al Shloul MA, Altakhaineh ARA, Saqer M. Role of activity-based learning and chatgpt on students' performance in education. *Educ Inform Technol.* 2024.
122. Guizani S, Mazhar T, Shahzad T, Ahmad W, Bibi A, Hamam H. A systematic literature review to implement large language model in higher education: issues and solutions. *Discovery Sustain.* 2025.
123. Shayea GG, Zabil MHM, Habeeb MA, Khaleel YL, Albahri AS. Strategies for protection against adversarial attacks in ai models: an in-depth review. *J Intell Syst.* 2025;34(1):20240277.